The Macro Transcension Hypothesis: Spinning Black Holes in Dense Stellar Clusters as Thermodynamic Attractors for Advanced Civilizations, with Omega Centauri as an Observational Test Bed

Tim Swanson
The Omega Centauri Society / Post Oak Labs
tim@postoaklabs.com

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Abstract

Most proposed resolutions of the Fermi paradox assume that long-lived technological civilizations either expand outward, perish, or deliberately hide. We develop a fourth alternative, the *Macro Transcension Hypothesis* (MTH): that civilizations which optimize for long-term computation are driven by thermodynamics, rather than preference, toward a specific class of astrophysical environment, namely rapidly spinning massive black holes embedded in dense, old stellar systems, and that the migration and its endpoint are both electromagnetically quiet. The MTH extends the transcension hypothesis of Smart (2012) from planet-scale "inner space" to macroscopic black-hole infrastructure, and differs from the aestivation hypothesis of Sandberg et al. (2016) in requiring no waiting strategy: the relevant free-energy and entropy-disposal advantages are available now. We quantify the case in four steps: (i) thin-disk accretion onto a Kerr black hole releases 5.7–30 per cent of rest-mass energy across the spins disk accretion can reach (Bardeen et al., 1972; Novikov and Thorne, 1973; Thorne, 1974), rising to 42 per cent only in the formal $a_\star \to 1$ limit that radiative spin equilibrium forbids, versus 0.7 per cent for hydrogen fusion, while magnetically arrested disks extract additional spin energy at effective efficiencies exceeding 100 per cent of accreted rest mass (Tchekhovskoy et al., 2011); (ii) by the generalized second law (Bekenstein, 1974), an event horizon is a thermodynamically ideal entropy sink, and a worked delivery budget (Section 4.2) shows the realized erasure cost is of order the CMB-limited Landauer floor (a gain of ~ 0.05 on the conservative loss budget) if the carrier modes are thermally occupied by the 2.7 K sky, and lands a factor of $\sim 10^3$ below it in a fed flow ($\sim 10^7$ in the unfuelled ambient medium) only if the modes are kept colder than the sky from the emitter to the horizon, which requires a sky-occluding aperture and a path not thermalized by plasma at $T_e \gg 2.7$ K; (iii) the Bekenstein–Hawking entropy of a $\sim 2 \times 10^4\ M_\odot$ black hole corresponds to $\sim 6 \times 10^{85}$ bits, or $\approx 3 \times 10^{85}$ at the Thorne spin the architecture ends at, exceeding any material archive; and (iv) per unit of harvested mass, this architecture outperforms complete fusion of the same fuel by a factor of $\sim$ 11–60, outperforms Dyson-type stellar harvesting without star lifting by $\sim$ 115–600 in lifetime energy yield ($\sim$ 11–60 against a star-lifting economy), and exceeds a solar Dyson swarm by a factor of $\sim 10^9$ (6.6×10^8 at $2 \times 10^4\ M_\odot$) in instantaneous Eddington-limited power.

We then identify Omega Centauri (NGC 5139), a $\sim 4\times10^6\,M_\odot$, ~12-Gyr-old stripped dwarf-galaxy nucleus hosting the nearest strong candidate intermediate-mass black hole (Häberle et al., 2024b), as the most observationally accessible system satisfying the MTH selection criteria, and present a falsification framework built on six instrument-matched tests: accretion-luminosity limits (JWST, ATCA), mid-infrared waste-heat limits (JWST), a gravitational-wave signature inconsistent with a single point mass, an EMRI point-mass measurement, and an EMRI spin measurement (the latter three via LISA, contingent on a compact-object inspiral occurring in-band during the mission), plus neutrino burst searches (KM3NeT). Current data, including the unresolved tension between a $\geq$ 8,200 $M_\odot$ kinematic lower bound (Häberle et al., 2024b) and a $\lesssim$ 6,000 $M_\odot$ pulsar-timing upper bound (Bañares-Hernández et al., 2025) and the complete electromagnetic silence of the central object (Mahida et al., 2026; Chen et al., 2026), are consistent with both the MTH and the more parsimonious gas-starvation null hypothesis; we state explicitly which forthcoming observations would discriminate between them, and which would falsify the MTH outright. The hypothesis is offered in the falsificationist tradition of Sandberg et al. (2016) and Dvali and Osmanov (2023): a speculative but physically grounded working model whose value lies in the concrete observational program it motivates.

Keywords: Fermi paradox; technosignatures; SETI; intermediate-mass black holes; Omega Centauri; NGC 5139; Blandford–Znajek mechanism; Landauer limit; transcension; aestivation

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1 Introduction

The Fermi paradox, in its modern form (Hart, 1975; Tipler, 1980; Webb, 2015; Ćirković, 2018; Forgan, 2019), rests on an expansionist premise: that at least some technological civilizations should grow outward across interstellar distances, and that such growth, sustained for even a small fraction of galactic history, would be conspicuous. The premise is quantitatively robust. Self-reproducing probe architectures permit galaxy-scale colonization in 10^6–10^8 years (Tipler, 1980; Freitas and Gilbreath, 1982), intergalactic expansion is energetically cheap for a mature civilization (Armstrong and Sandberg, 2013), and surveys for the waste heat of large-scale energy harvesting have returned null results across $\sim 10^5$ galaxies (Wright et al., 2014; Griffith et al., 2015). The observed silence therefore demands either that technological life is extremely rare, that it reliably destroys itself, or that the expansionist premise is wrong.

A minority tradition in the literature attacks the premise itself. Smart (2012) proposed the *transcension hypothesis*: that the developmental trajectory of intelligence points not outward but inward, toward ever denser, faster, and more efficient configurations of matter and energy ("STEM compression" of space, time, energy, and matter), terminating, in Smart's telling, at black-hole-scale densities and an effective exit from the observable universe. Sandberg et al. (2016) proposed the *aestivation hypothesis*: that civilizations which value computation should defer it to the far future, when the cosmic microwave background (CMB) has cooled and each erased bit costs less free energy, and should therefore be dormant now. Both proposals share a key insight, that the thermodynamics of computation is the correct lens through which to predict the behaviour of mature intelligence, and both have been criticized on physical grounds. Smart's mechanism for leaving the universe is not specified in testable form, and the aestivation argument was answered by Bennett et al. (2019), who showed that erasure can be paid at any epoch and is not improved by waiting for a colder background, so dormancy confers no advantage. The earliest proposal found that black holes could host intelligence predates both: Lefebvre and Efremov (2000) argued that black holes may serve as a physical substratum for intelligent beings, on the strength of structural analogies between Kerr black holes and the brain–psyche system. The grounding there is analogy rather than any energy logic, and the thermodynamic argument developed here is independent of it.

This paper develops a third member of that family, which we call the *Macro Transcension Hypothesis* (MTH). Its core claim is that the inward trajectory identified by Smart has a concrete, macroscopic, observationally accessible destination in ordinary astrophysics: rapidly spinning massive black holes embedded in dense, dynamically old stellar systems. No new physics, no exit from the universe, and no multi-gigayear dormancy are required. The advantages that drive the migration (accretion power at up to ∼42 times the mass-energy efficiency of hydrogen fusion at the highest spin disk accretion reaches, an event horizon serving as an entropy sink of unlimited capacity, information storage at the Bekenstein–Hawking density, and gravitational time dilation as a controllable computational resource) are all available in the present epoch, and all follow from textbook general relativity and black-hole thermodynamics (Misner et al., 1973; Bardeen et al., 1972; Bekenstein, 1973, 1974). The hypothesis predicts that the most advanced civilizations remain present but electromagnetically invisible, because maximum computational efficiency, viewed from outside, is electromagnetically quiet.

The MTH would remain idle philosophy if it did not select targets. The selection criteria (a massive black hole of modest depth, intermediate rather than supermassive; a dense retinue of low-mass stars as fuel; dynamical age sufficient for any migrating civilization to have arrived and matured; and a quiescent electromagnetic environment) converge on a short list of Galactic systems, at the top of which sits Omega Centauri (ω Cen, NGC 5139): the Milky Way's most massive globular cluster, widely interpreted as the stripped nucleus of an accreted dwarf galaxy (Hilker and Richtler, 2000; Bekki and Tsujimoto, 2019; Ibata et al., 2019), host to $\sim 10^7$ stars of mean age 12.08 ± 0.01 Gyr (Clontz et al., 2024), and, since the detection of seven fast-moving stars within its central 3 arcsec (five of which are robust detections), host to

the strongest current candidate for an intermediate-mass black hole (IMBH) in the Galaxy (Häberle et al., 2024b). That candidate is simultaneously the subject of a sharp observational controversy: combined stellar-kinematic and millisecond-pulsar-timing analysis places a 3σ upper limit of $\sim$6,000 $M_\odot$ on any central point mass (Bañares-Hernández et al., 2025), formally inconsistent with the kinematic lower bound. And it is electromagnetically dark to remarkable depth: neither $\sim$170 hours of ATCA radio integration (Mahida et al., 2026), nor JWST infrared photometry (Chen et al., 2026), nor a $\sim$291 ks Chandra exposure (Haggard et al., 2013) detects any accretion signature. Every element of this situation (a contested central mass, perfect electromagnetic silence, an imminent decisive instrument in LISA) makes ω Cen a natural laboratory for testing inward-migration hypotheses, entirely independent of whether the MTH is true.

1.1 Claims

Because the argument mixes established physics with speculative inference, we state the epistemic register of each claim explicitly. The paper makes three claims of different strength:

1. **(Established physics plus a comparative synthesis.)** Spinning black holes in dense old clusters are, on the comparison set stated here, the most thermodynamically favourable present-day environments we identify for long-term, large-scale computation, by $\sim$11–60 in energy per unit fuel against complete fusion. Whether a maximizer prefers them to a cold-space architecture is left open (Section 4.2). The ingredient results are established physics; the ranking is a synthesis over a stated comparison set (fusion and Dyson-type stellar harvesting; cold-space computing is compared in Section 4.2 without a ranking), defended quantitatively in Section 4 using standard results (Novikov and Thorne, 1973; Blandford and Znajek, 1977; Bekenstein, 1974; Landauer, 1961; Lloyd, 2000), and is independent of any claim about extraterrestrial intelligence. The ranking of cluster IMBHs above supermassive holes rests partly on strategic rather than thermodynamic criteria (hazard exposure, contested resources), which we treat separately in Section 5.5.

2. **(Speculative hypothesis.)** If long-lived technological civilizations exist and even a subset optimizes for computational capacity (Sandberg, 1999), the environments of claim 1 act as attractors[1], and the resulting migration resolves the Fermi paradox without rare-Earth or doomsday assumptions. This is the MTH proper, stated formally in Section 3. We regard it as less parsimonious than the null hypothesis (technological life is rare or absent) and say so.

3. **(Observational program.)** Whether or not claim 2 is true, it generates instrument-matched, time-bounded, falsifiable predictions at ω Cen, several of which will be adjudicated by existing or funded instruments within $\sim$15 years. This program is presented in Section 6.

We do not claim that ω Cen hosts a civilization; we do not claim the IMBH is confirmed (it is not); and we do not claim the observed electromagnetic silence is evidence for the MTH, since it is equally consistent with the simpler gas-starvation explanation that we adopt as the null hypothesis throughout.

1.2 Structure of the paper

Section 2 situates the MTH in the literature. Section 3 states the hypothesis formally as four postulates. Section 4 develops the quantitative thermodynamic and computational case. Section 5 reviews ω Cen and

[1]The attractor vocabulary has recent independent precedent: Kriger (2026), a Zenodo preprint of 31 January 2026, frames the inward turn of advanced civilizations as computational efficiency acting as a universal attractor, offered as a Fermi Paradox solution and derived from the energy-cost asymmetry between moving mass and processing information. It makes no black-hole or cluster-specific claims and selects no astrophysical environment.

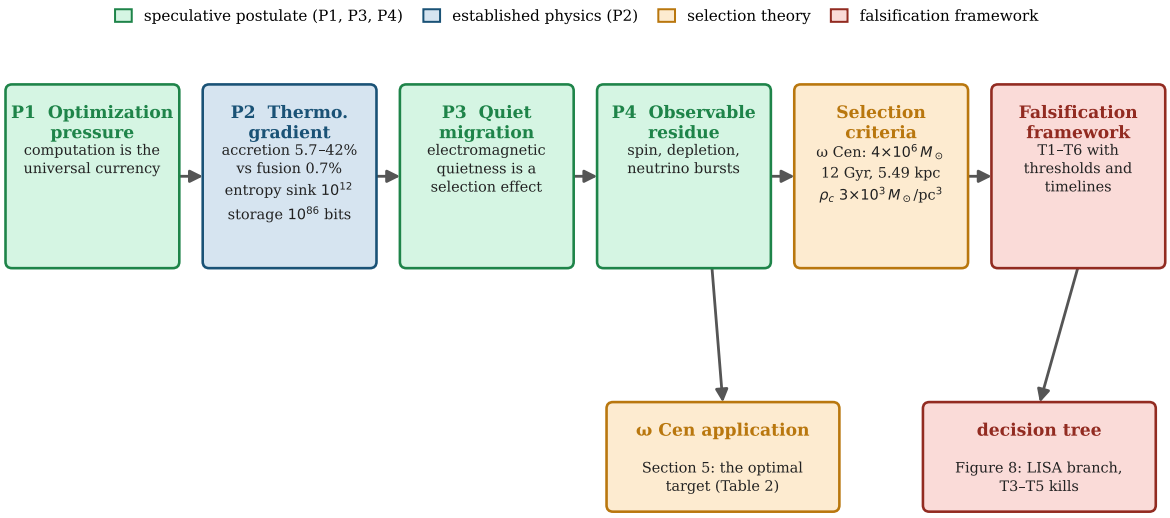


Figure 1: The argument chain of the paper as a left-to-right cascade, coloured by the epistemic register of Section 1.1: green, the speculative postulates (P1, P3, P4); blue, the established physics of P2 (Section 4); orange, the selection theory (Section 3.1); red, the falsification framework (Section 6, Table 7). Each box carries the quantitative anchor developed in the section it points to. The observable residue P4 branches to the target application (Section 5), and the framework terminates in the decision tree of Figure 7.

its IMBH evidence, including the current mass tension. Section 6 presents the falsification framework and observational roadmap; the staged architecture that motivates it, sketched at the level of explicitly labelled speculation, is developed in Appendix A. Section 7 addresses objections, and Section 8 concludes. Figure 1 previews the argument chain and its epistemic register.

2 Relation to prior work

2.1 The expansionist mainstream

The canonical statements of the Fermi paradox (Hart, 1975; Tipler, 1980) treat interstellar expansion as the default behaviour of technological life, an assumption sharpened by Armstrong and Sandberg (2013), who showed that a single mature civilization could seed every galaxy within several billion light-years using a fraction of one star's output. The "grabby aliens" model of Hanson et al. (2021) quantifies the selection effect: if loud (expanding) civilizations exist at any appreciable rate, they dominate the future light cone, and our own early arrival is informative about their density. Observational programs built on the expansionist premise (radio and optical SETI, Tarter 2001; waste-heat surveys, Dyson 1960; Wright et al. 2014; Griffith et al. 2015; and the broader technosignature framework, Wright et al. 2022; Socas-Navarro et al. 2021; Lingam and Loeb 2021) have produced consistent null results. Within the expansionist frame, these nulls force the unattractive trilemma of rarity, doom, or concealment.

2.2 The inward-migration family

The MTH belongs to a smaller family of proposals holding that mature intelligence migrates down thermodynamic gradients rather than out across space.

Transcension. Smart (2012) argued that the developmental history of complexity on Earth exhibits sustained compression in space, time, energy, and matter ("STEM compression"), and extrapolated to a future in which civilizations engineer environments of black-hole density and disappear from the observable universe. The MTH adopts the compression premise but rejects the terminal step: nothing in known physics permits or requires an exit from the universe, and a hypothesis terminating in unobservable territory cannot be tested. The MTH instead halts the inward trajectory at a macroscopic astrophysical object, an existing massive black hole, where every claimed advantage is calculable and several are observable, which is the sense in which the transcension here is macro.

Postbiological migration. The closest precursor to the present proposal is Ćirković and Bradbury (2006), who argued that postbiological civilizations should migrate down Galactic thermodynamic gradients toward the cold outer Galaxy, where computation is cheaper, and that SETI's apparent failure reflects a search aimed at the wrong locations. The MTH shares the thermodynamic-migration logic and the diagnosis of SETI's targeting problem, but identifies a different optimum: the relevant gradient, we argue, is set by the conjunction of energy density, entropy-disposal capacity, and storage density rather than by ambient temperature alone, and its maxima are spinning black holes in dense old clusters rather than the Galactic outskirts.

Aestivation. Sandberg et al. (2016) proposed that civilizations defer computation until the CMB cools, gaining a factor of up to $\sim 10^{30}$ in computations per joule, and so should currently be dormant. Bennett et al. (2019) showed that the erasure cost aestivation defers can be paid at any epoch, by transferring entropy into existing non-maximal-entropy reservoirs, and that this is not improved by waiting for a colder background, so dormancy confers no advantage. A separate consideration, ours, is that stars burn whether or not anyone harvests them, so maximizers should be active now. The MTH avoids the aestivation trade-off by construction: instead of waiting for a colder universe, it imports a colder-than-CMB entropy sink into the present. Waste entropy crossing an event horizon increases horizon area in accordance with the generalized second law (Bekenstein, 1974), at an effective marginal temperature far below the CMB's 2.7 K (Section 4.2). An MTH civilization harvests aggressively in the present epoch, consistent with the Bennett et al. (2019) result that waiting does not lower the erasure cost, while enjoying much of the efficiency gain that motivated aestivation.

Black holes as civilization infrastructure. The conceit is older than SETI's engagement with it. Penrose (1969) introduced rotational-energy extraction with an advanced-civilization framing, and Misner et al. (1973, §33.7) formalized it as the garbage-dump thought experiment: a civilization lowers its waste to just outside the horizon, releases it, and recovers up to 29 per cent of the rest mass as work. Thorne (1994) staged the same power station for a popular audience, and Frolov and Novikov (1998, §12) give the textbook treatment of black holes as energy sources for civilizations. The present series is a completion of that thread. Within SETI proper, several authors have treated black holes as energy sources or computational substrates for advanced intelligence: Inoue and Yokoo (2011) proposed Dyson-type collectors around an accreting supermassive black hole; Hsiao et al. (2021) computed the detectability of a "Dyson sphere around a black hole" harvesting disk luminosity; Opatrný et al. (2017) analysed the inverted configuration in which a black hole serves as the cold reservoir of a heat engine run against the warmer sky; Vidal (2011, 2014) argued that black-hole-coupled ("stellivore") systems are the natural attractor for cosmic intelligence; and Dvali and Osmanov (2023) analysed black holes as quantum information processors for extraterrestrial civilizations, predicting characteristic high-energy neutrino emission from artificial micro-black-hole computers. Lacki and DiKerby (2025) survey the corresponding high-energy SETI opportunity, and the Dyson Minds workshop (Curtis et al., 2026) has since brought this thread into mainstream observational planning, recommending anomaly-detection searches of archival WISE, JWST, and EHT data for black-hole-hosted intelligence, with a focus on supermassive hosts. The MTH

synthesizes this thread and adds the two elements it has lacked: a *selection theory* specifying which black holes are preferred and why (intermediate-mass, high-spin, cluster-embedded; Section 3.1), and a named, ranked observational target with a costed test program.

Kardashev and his inversion. The Kardashev scale (Kardashev, 1964) ranks civilizations by total energy throughput, implicitly rewarding outward expansion. Barrow (1998) proposed the complementary inward scale: mastery over ever-smaller scales of structure. The MTH is the astrophysical completion of Barrow's inversion — the observation that the universe already provides, free of charge, the densest stable structures that inward mastery could aim at, and that the rational endpoint of a Barrow-type trajectory is therefore relocation rather than fabrication. Bradbury (1999) reached a related conclusion from the engineering side: his Matrioshka-brain analysis shows stellar-powered computation is ultimately limited by radiator area and sink temperature, the constraints an event horizon eliminates.

2.3 New content of the hypothesis

To our knowledge this paper is the first to (i) state the inward-migration resolution of the Fermi paradox as a falsifiable selection theory over real astrophysical objects; (ii) integrate the generalized second law, the Landauer bound, Kerr accretion efficiency, Blandford–Znajek extraction, and Bekenstein–Hawking storage into a single quantitative budget for a civilization-scale computer hosted by an existing black hole; and (iii) attach the resulting predictions to a specific system (ω Cen) with instrument-matched falsification thresholds and timelines. Each ingredient exists in the literature; the synthesis, the selection criteria, and the test program are new.

3 The hypothesis

Epistemic status: Sections 3–4 contain, respectively, a speculative hypothesis stated formally, and established physics offered in its support. The physics in Section 4 stands regardless of the hypothesis.

We state the MTH as four postulates.

P1 (Optimization pressure). Among long-lived technological civilizations, a non-empty subset evolves (by competition, self-modification, or design) toward maximizing long-term computational capacity per unit of controlled resource, under at least one of three conditions on the objective: (i) temporal discounting, (ii) a requirement that the computation be unified and latency-bound, or (iii) a bounded planning horizon. The three are needed because an undiscounted maximizer of total integrated computation over unbounded time is served by expansion, for reasons set out in Section 4.5; they are stated here rather than added later. (Motivation: computation is the universal currency into which information-processing agents can convert essentially any terminal goal. This is the instrumental-convergence thesis (Omohundro, 2008; Bostrom, 2012), given formal treatment by Benson-Tilsen and Soares (2016), and the same premise adopted by Smart 2012 and Sandberg et al. 2016. What remains open is the step from convergent instrumental goals in a single agent to a population claim over lineages; P1 asserts only a non-empty subset, the weakest form that step permits.)

P2 (Thermodynamic gradient). For any such maximizer, the present-day universe contains a strict gradient of substrate quality whose optima over the comparison set of Section 4 are rapidly spinning massive black holes embedded in dense, old, low-luminosity stellar systems. The gradient is set by four independent physical quantities: extractable energy per unit fuel mass, entropy-disposal

capacity, information-storage density, and control over effective clock rate. (Defended quantitatively in Section 4.)

P3 (Quiet migration). Migration toward and exploitation of such environments is electromagnetically inconspicuous at interstellar distances. Transit uses gram-to-kilogram-scale beamed-sail payloads rather than worldships (Lubin, 2016); construction uses in-situ resources (Freitas and Gilbreath, 1982); and mature operation, by the very efficiency that motivates it (P2), radiates negligible waste heat (Section 4.2).

P4 (Observable residue). The endpoint is nevertheless not perfectly unobservable. It predicts (a) anomalously high spin on the host black hole; (b) suppressed or absent accretion luminosity relative to ambient gas supply; (c) depletion of the loosely bound stellar population over Gyr timescales, observable today as a low-mass stellar mass-function slope in the central ∼0.1 pc falling below the envelope of collisional N-body models without engineered feeding (González Prieto et al., 2025); and (d) possibly, burst-mode high-energy neutrino emission if micro-black-hole computation of the Dvali and Osmanov (2023) type is employed. These residues define the falsification program of Section 6.

The hypothesis (MTH). *If P1–P3 hold, and migration is achievable at a cost the gradient repays (Section 4.5), then the oldest computation-maximizing civilizations in the Galaxy have already relocated to environments of the P2 class; their absence from planetary systems, the radio spectrum, and waste-heat surveys is a selection effect of the thermodynamic gradient; and the appropriate search strategy is targeted scrutiny of the small set of P2-optimal Galactic systems for the P4 residues.*

Three corollaries sharpen the contrast with neighbouring hypotheses. First, against aestivation: the MTH predicts present activity in P2 environments, not dormancy, so it does not incur the dormancy that Bennett et al. (2019) show confers no advantage. Second, against the dark-forest family of concealment hypotheses: silence here is a by-product of efficiency, not a strategic choice, so the MTH requires no game-theoretic coordination among civilizations. Third, against Smart's original transcension: every MTH process occurs in ordinary spacetime around an ordinary Kerr black hole, so the hypothesis remains permanently coupled to observation.

3.1 Selection criteria over host systems

P2 implies a ranking of real systems. The optimum is set by competing requirements:

1. **Black-hole mass in the intermediate range (10^3–$10^5\ M_\odot$).** Energy extraction (Section 4.1) scales with available fuel and with Eddington-limited throughput ($\propto M$), favouring larger M. The lower bound is set by tides. The differential field at the Schwarzschild ISCO is $2GM/r_{\rm ISCO}^3 = c^6/(108\,G^2M^2)$, which is $\simeq 0.95\ \mathrm{s}^{-2}$ at $2\times10^4\ M_\odot$ and scales as M^{-2}; requiring a 10-m free-flying node to stay below $\sim10^2\ g$ of differential acceleration gives $M \gtrsim 2\times10^3\ M_\odot$, of which the stated 10^3 floor is the round order of magnitude. The upper bound is not a cost that grows with M but two considerations that cap the useful range: Bekenstein–Hawking storage already saturates any plausible need at $10^4\ M_\odot$ (Section 4.3), so further mass buys nothing on that axis, and the environments in which holes above $10^5\ M_\odot$ are actually found are deep galactic-centre potentials crowded with disruptive astrophysics (Section 5.5). The intermediate range secures every advantage at minimal exposure.

2. **High spin, or spin-up feasibility.** The Blandford–Znajek channel and the deep-ISCO efficiencies require $a_\star \gtrsim 0.9$; a hole of modest spin can be spun up by prograde disk accretion at a calculable mass cost (Thorne, 1974), provided fuel exists (criterion 3).

3. **A dense, old, low-mass stellar reservoir.** Sustained Eddington-limited feeding of a $\sim 2 \times 10^4\, M_\odot$ hole consumes of order one solar mass per two millennia (Section 4.1); at ω Cen's central density the inner parsec supplies 3.7×10^4 stars and the central ~10 pc of order 4.3×10^6 (Table 5), at relative velocities of tens of $\mathrm{km\,s^{-1}}$. The contested status of this fuel is era-relative: it is uncontested by fusion-era economics, but under star lifting the same M dwarfs are prize fuel (fully convective, hence hydrogen-complete, with $\sim 10^{12}$-yr lifetimes; Scoggins and Kipping 2023), so among computation-maximizing lineages the reservoir is maximally contested, an early-arrival premium developed in the companion economics paper.

4. **Dynamical age and quiescence.** A system $\gtrsim$10 Gyr old has been reachable by any Galactic civilization arising in the first stellar generations; a relaxed, gas-poor environment minimizes both natural accretion noise and hazards to infrastructure.

Milky Way globular clusters with massive-IMBH candidates satisfy all four criteria simultaneously; ω Cen satisfies them maximally (Section 5). The Galactic Centre fails criterion 1 and 4; isolated stellar-mass black holes fail criterion 3; young massive clusters fail criterion 4. Retention is not the bottleneck: survival analyses of IMBHs in dense clusters find that holes above $\sim 10^3\, M_\odot$ are generally retained against dynamical ejection over a Hubble time (Martinez et al., 2026), so if formation occurs at all, the P2 class is populated today. Figure 2 locates the measured-spin stellar-mass holes, the IMBH candidates, and the P2 selection region in the mass–spin plane.

Because a selection theory that never exhibits its selection invites the charge of unfalsifiability by retreat, Table 1 scores the leading Galactic candidates against the four criteria. The ranking is coarse (the IMBH-evidence column in particular is contested for every entry), but it is the operational content of P2: the class-level kill statement of Section 7 quantifies over exactly these systems.

4 The thermodynamic and computational case

This section defends postulate P2 quantitatively. Throughout, M is the black-hole mass, $a_\star \equiv Jc/GM^2 \in [0, 1)$ the dimensionless spin, $r_g \equiv GM/c^2$ the gravitational radius, and we evaluate fiducial numbers at $M = 2 \times 10^4\, M_\odot$, the geometric midpoint of the kinematic side of the contested ω Cen range, $\sqrt{8.2 \times 10^3 \times 5 \times 10^4} = 2.0 \times 10^4$ (Section 5); the pulsar-timing side would place the midpoint lower, and we use the kinematic interval throughout so that the fiducial is stated rather than selected. All results in this section are standard physics; no claim about extraterrestrial intelligence is made until Appendix A.

4.1 Energy: accretion and spin extraction versus fusion

Thin-disk efficiency. Matter spiralling through a geometrically thin, radiatively efficient disk is released at the ISCO binding energy. For a circular equatorial orbit at radius r around a Kerr hole, the specific energy at the ISCO gives a radiative efficiency (Bardeen et al., 1972; Novikov and Thorne, 1973)

$$\eta(a_\star) \;=\; 1 - E_{\mathrm{ISCO}} \;=\; 1 - \sqrt{1 - \frac{2}{3}\,\frac{r_g}{r_{\mathrm{ISCO}}(a_\star)}}, \tag{1}$$

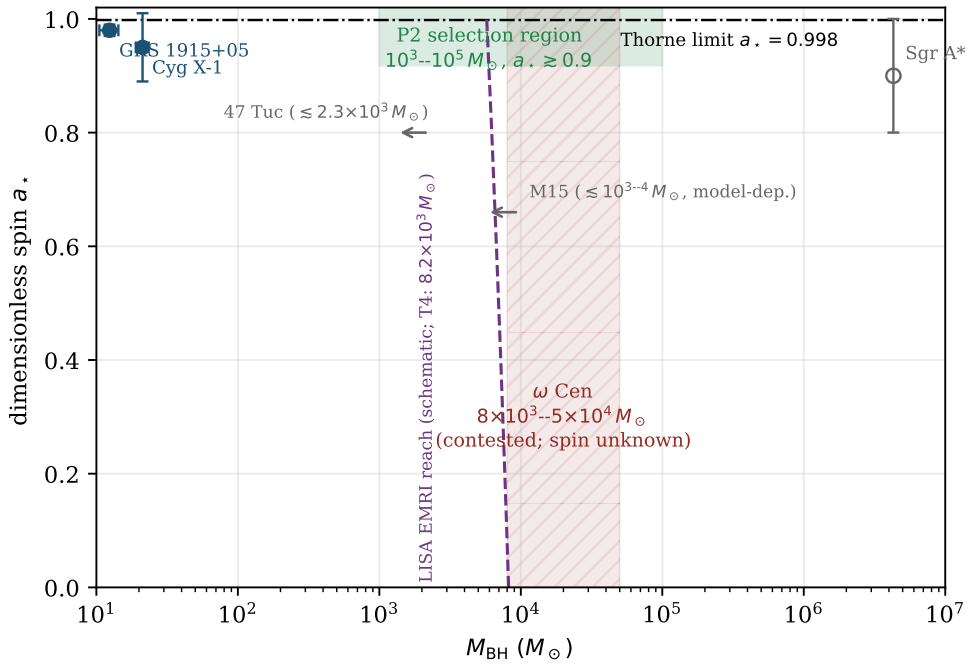


Figure 2: The mass–spin plane. Blue: stellar-mass black holes with spin measurements (representative continuum-fitting values; compilation in Reynolds 2021). Red hatched band: the contested ω Cen central mass, 8×10^3–5×10^4 $M_\odot$ (Table 6), spin unknown. Grey arrows: IMBH candidates constrained only from above (47 Tuc, Kızıltan et al. 2017; M15, Greene et al. 2020). Grey open symbol: Sgr A* (spin inferred; GRAVITY Collaboration 2022). Green: the P2 selection region, 10^3–10^5 $M_\odot$ at $a_\star \gtrsim 0.9$ (criteria 1–2). Dash-dotted line: the Thorne (1974) spin-equilibrium limit $a_\star = 0.998$. Dashed curve: schematic LISA EMRI mass reach at 5.49 kpc, anchored at the T4 threshold of 8.2×10^3 $M_\odot$ (Table 7) and drawn slightly deeper at high spin, where the inspiral spends more cycles in band. The curve states a requirement the test places on LISA, not a demonstrated reach at this distance; no mission-level sensitivity study for cluster EMRIs at 5.49 kpc is cited here, and the threshold should be read as the mass resolution T4 would need.

Table 1: The P2 selection criteria applied to the leading Galactic candidate systems. Symbol columns (current IMBH evidence; inferred or claimed mass versus the preferred 10^3–$10^5\,M_\odot$ range; low-mass fuel reservoir; dynamical age and quiescence): ++ strong, + favourable, ∘ neutral or contested, − unfavourable. Quantitative columns: distance and central mass density from Harris (1996); Baumgardt and Vasiliev (2021), rounded to leading digit; $r_{\rm infl} = GM/\sigma_c^2$ evaluated at a reference $10^4\,M_\odot$ and the catalogued central dispersion, which for ω Cen is $\sigma_c = 16.8\,{\rm km\,s^{-1}}$ (Harris, 1996) and gives 0.15 pc; the 0.10 pc printed for this row in earlier versions corresponded to $\sigma_c = 20.7\,{\rm km\,s^{-1}}$ and is corrected (Appendix C). Values assume an isotropic, spherical central-dispersion estimate and carry ∼50 per cent systematics from that assumption alone; best current central-mass constraint with source.

System	Ev.	Mass	Fuel	Age	d (kpc)	ρ_c ($M_\odot\,{\rm pc^{-3}}$)	$r_{\rm infl}$ (pc)	Best current mass constraint ($M_\odot$)
ω Cen	++	++	++	++	5.5	$\sim 3\times10^3$	0.15	$\geq 8.2\times10^3$ (kinematic) vs. $< 6\times10^3$ (timing); contested (Section 5.3) (Häberle et al., 2024b; Bañares-Hernández et al., 2025)
47 Tuc	∘	+	++	++	4.5	$\sim 2\times10^5$	0.3	$\sim 2.3\times10^3$ timing claim (Kızıltan et al., 2017); not required by later modelling (Mann et al., 2019)
M54	+	+	+	+	26.3	$\sim 10^5$	0.2	$\sim 10^4$ kinematic hint (Greene et al., 2020); Sgr dwarf nucleus
NGC 6388	∘	+	+	+	9.9	$\sim 5\times10^5$	0.12	kinematic claims disputed; deep radio null, $\lesssim$ few $\times 10^3$ (Tremou et al., 2018)
NGC 6441	∘	+	+	+	11.6	$\sim 4\times10^5$	0.13	as NGC 6388; radio null (Tremou et al., 2018)
Terzan 5	−	∘	++	∘	6.9	$\sim 3\times10^5$	0.2	no direct evidence (heavy bulge extinction); radio null (Tremou et al., 2018)
M15	∘	∘	++	++	10.4	$\sim 10^{6–7}$ (cusp)	0.25	$\lesssim 10^{3–4}$, model-dependent; kinematics equally fit by remnants (Greene et al., 2020)

which rises from $\eta = 1 - \sqrt{8/9} \simeq 0.057$ at $a_\star = 0$ ($r_{\rm ISCO} = 6\,r_g$) to 0.321 (pure geodesic value; the photon-capture-corrected equilibrium efficiency is $\simeq 0.30$) at the Thorne (1974) radiative spin-equilibrium limit $a_\star = 0.998$, and formally to $1 - 1/\sqrt{3} \simeq 0.423$ as $a_\star \to 1$ (Fig. 3). Hydrogen fusion releases 0.71 per cent of the rest mass burned, but a real star offers only its hydrogen fraction $X_{\rm H} \simeq 0.7$ to that channel, so complete fusion of a star yields $\simeq 5\times10^{-3}$ of stellar mass; an unassisted main-sequence star processes roughly a tenth of its mass through its core over its lifetime, and only the hydrogen in that tenth burns, so a Dyson-type collector (Dyson, 1960) captures a lifetime yield of order 5×10^{-4}. Both denominators therefore carry $X_{\rm H}$. Per unit of fuel mass under the civilization's control, disk accretion onto a high-spin hole outperforms complete fusion of the same mass by a factor of ∼11 at $a_\star = 0$ and ∼60 at the Thorne cap (formally ∼85 as $a_\star \to 1$), and outperforms Dyson harvesting without star lifting by ∼115–600 (formally ∼850). Under a hydrogen-only convention for the denominators the Dyson factors would read ∼80–420 instead; the numerator is the same, and only the treatment of $X_{\rm H}$ differs (Appendix C). The comparison set matters, because star lifting changes it: a civilization that strips the envelope and feeds the core accesses close to the full hydrogen inventory (Criswell, 1985; Matloff, 2017; Scoggins and Kipping, 2023), raising the yield toward the 5×10^{-3} ceiling and collapsing the Dyson disadvantage to the same ∼11–60 as complete fusion. The entropy-sink and storage advantages (Sections 4.2–4.3) are independent of which factor applies.

Power throughput. The sustainable luminosity is bounded by the Eddington limit,

$$L_{\rm Edd} = \frac{4\pi G M m_p c}{\sigma_T} \simeq 1.26\times10^{31}\left(\frac{M}{M_\odot}\right)\,{\rm W}, \tag{2}$$

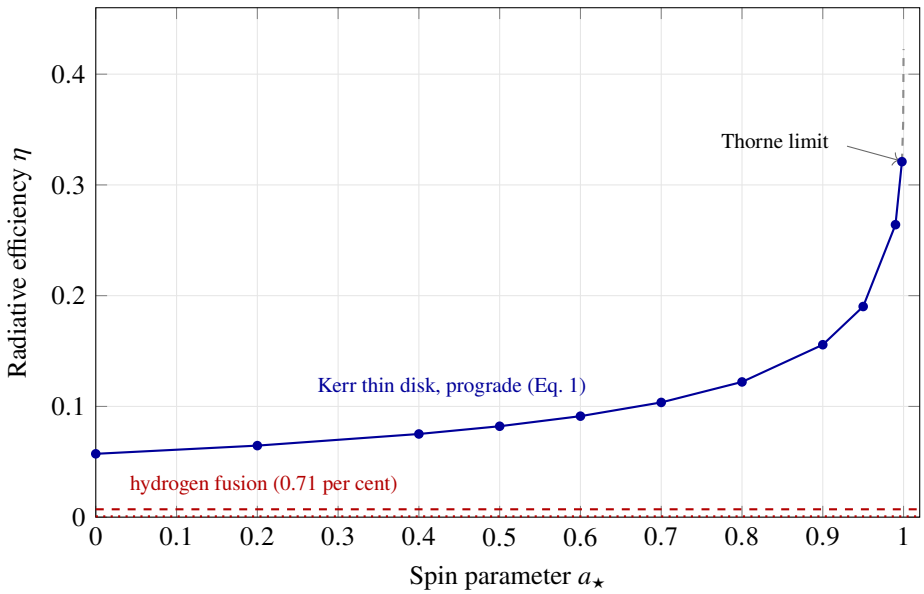


Figure 3: Mass-to-energy conversion efficiency of thin-disk accretion as a function of black-hole spin (Bardeen et al., 1972; Novikov and Thorne, 1973), compared with hydrogen fusion and with the lifetime yield of Dyson-type stellar harvesting. The Thorne (1974) limit $a_\star = 0.998$ marks the equilibrium spin reachable by disk accretion alone; the plotted $\eta \simeq 0.32$ there is the geodesic value, reduced to $\simeq 0.30$ once photon capture is included (Thorne, 1974). Curve drawn through exact values of Eq. (1) at the plotted points; the solid branch terminates at the Thorne cap, and the grey dashed continuation to $a_\star \to 1$ marks the formal limit, which radiative spin equilibrium places out of reach.

giving 1.0×10^{35}, 2.5×10^{35}, and 6.3×10^{35} W at $M = 8.2 \times 10^3$, 2×10^4, and $5 \times 10^4\, M_\odot$ respectively (Frank et al., 2002; Shapiro and Teukolsky, 1983), some 3×10^8–2×10^9 times the bolometric output of the Sun across that mass range (6.6×10^8 at $2 \times 10^4\, M_\odot$), and hence of any solar Dyson swarm (Fig. 4). The corresponding Eddington accretion rate, $\dot{M}_{\rm Edd} = L_{\rm Edd}/\eta c^2$, is remarkably modest: at $\eta = 0.1$ and $M = 2 \times 10^4\, M_\odot$, one solar mass sustains the limit for 2,250 years. At the central density of this paper's King model ($\rho_c = 4.0 \times 10^3\, M_\odot\,{\rm pc}^{-3}$; Section 5.2) the central parsec holds $1.6 \times 10^4\, M_\odot$, 3.7×10^4 stars, or 3.6×10^7 yr of fuel at fixed hole mass; the working reservoir is the central ~10 pc, whose 4.3×10^6 stars amount to 4.2×10^9 yr on the same fixed-mass convention, or 2.0×10^8 yr of elapsed operation once the growing Eddington limit is accounted (Table 5) using bodies (brown dwarfs, low-mass M dwarfs, white dwarfs) that no fusion-based economy values, at the delivery cost priced in Appendix A.

Spin energy and the Blandford–Znajek channel. A Kerr hole stores extractable rotational energy

$$E_{\rm rot} = \left[1 - \sqrt{\tfrac{1}{2}\left(1 + \sqrt{1 - a_\star^2}\right)}\,\right] Mc^2, \tag{3}$$

reaching $0.29\, Mc^2 \simeq 1.0 \times 10^{51}$ J for $M = 2 \times 10^4\, M_\odot$ at $a_\star \to 1$: a reservoir equal to the entire lifetime fusion output of ~10^7 Suns, storable indefinitely and tappable on demand. The bracketed quantity is the Christodoulou–Ruffini reducible energy, the difference between M and the irreducible mass $M_{\rm irr}$ (Christodoulou, 1970; Christodoulou and Ruffini, 1971); extraction is reversible only in the limit of vanishing horizon-area increase, and inefficient extraction raises $M_{\rm irr}$ permanently, destroying future capacity, a distinction that matters for the spin-fossil reasoning of Section 6. That this energy is extractable in principle was shown by the ergosphere mechanism of Penrose (1969) and Penrose and Floyd (1971);

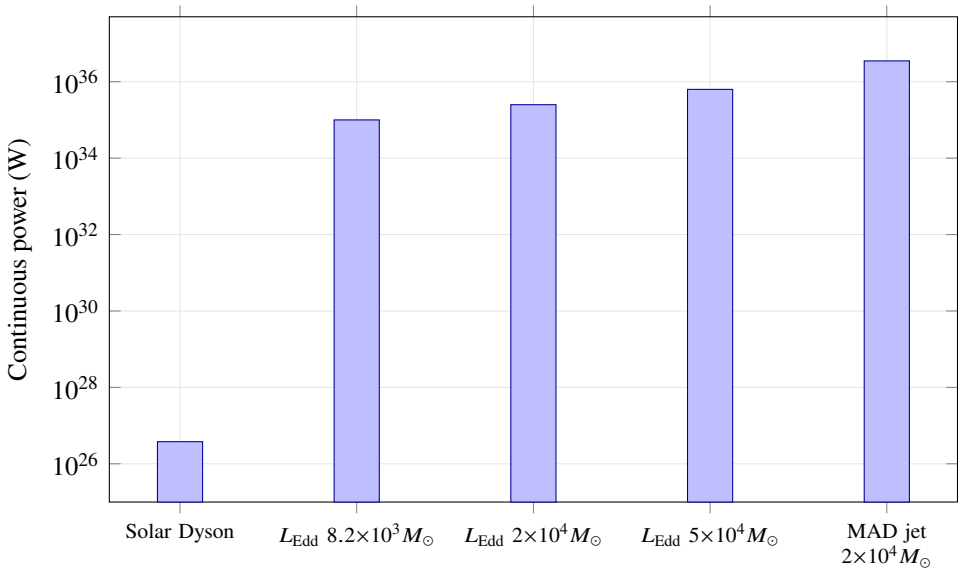


Figure 4: Continuous power available to a civilization from a complete solar Dyson swarm versus Eddington-limited accretion (Eq. 2) onto IMBHs spanning the contested ω Cen mass range, and a magnetically arrested (MAD) Blandford–Znajek jet at $a_\star \simeq 0.99$ with $\eta_{\rm jet} \simeq 1.4$ (Tchekhovskoy et al., 2011), fed at the $\eta = 0.1$ Eddington rate; the two efficiency conventions combine as $P_{\rm jet} = \eta_{\rm jet}\,\dot{M}c^2 = \eta_{\rm jet}\,(L_{\rm Edd}/\eta)$. The IMBH advantage in instantaneous power is roughly nine orders of magnitude.

the astrophysically efficient extraction mechanism is electromagnetic: a horizon threaded by poloidal magnetic flux Φ supplied by an accretion disk acts as a unipolar inductor, driving Poynting-flux jets with power $P_{\rm BZ} \propto \Phi^2\Omega_H^2$, where Ω_H is the horizon angular frequency (Blandford and Znajek, 1977). In the membrane paradigm of Thorne et al. (1986) the horizon carries a surface resistivity of 377 Ω and the whole configuration reads as a battery with internal resistance, the engineering-facing form of the mechanism and the one relevant to Appendix A's power-station framing. In the magnetically arrested disk (MAD) regime, general-relativistic magnetohydrodynamic simulations find jet efficiencies $\eta_{\rm jet} \equiv P_{\rm jet}/\dot{M}c^2 \simeq 1.4$ at $a_\star \simeq 0.99$ (Tchekhovskoy et al., 2011; Tchekhovskoy, 2015): the jet carries more energy than the accreted rest mass, the excess drawn from spin. The BZ scaling has recently been confirmed from first principles by ab-initio particle-in-cell simulations of Kerr magnetospheres (Meringolo et al., 2025), which additionally identify equatorial magnetic reconnection as a supplementary extraction channel (Comisso and Asenjo, 2021; Camilloni and Rezzolla, 2025); one 2026 preprint reports transient MAD states with jet power exceeding accretion power by over two orders of magnitude (Nathanail, 2026, not yet peer-reviewed). For the MTH it suffices that $\eta_{\rm jet} \gtrsim 1$ is robust across methods: an engineered MAD configuration around a spun-up IMBH is, per unit fuel, the most efficient large-scale power plant permitted by demonstrated physics.

Spin-up economics. A hole of low natal spin is itself improvable. Prograde thin-disk accretion drives $a_\star \to 0.998$ after the hole grows by a factor ≈ 2.2 (2.2024) (Bardeen, 1970; Thorne, 1974); for an initial $2 \times 10^4\ M_\odot$ hole this raises the hole mass by $2.4 \times 10^4\ M_\odot$, which is $2.91 \times 10^4\ M_\odot$ of delivered rest mass once the ISCO binding energy is charged along the track ($\mathrm{d}M_{\rm hole} = E_{\rm ISCO}\,\mathrm{d}M_{\rm rest}$, giving a ratio of 1.209), and consumes 6.8×10^4 cluster stars at $0.43\,M_\odot$. The percentages of Appendix A and the tug masses below are keyed to the hole increment, which is the quantity that leaves the reservoir as bound stellar mass minus what the disk radiates away. At Eddington throughput the episode runs $\sim 3.4 \times 10^7$ yr with the

rate evaluated at the mean hole mass during the episode. Integrating instead against the rising efficiency and the rising Eddington limit together, $\mathrm{d}t = \eta(a_\star)c^2\,\mathrm{d}M/L_{\mathrm{Edd}}(M)$, gives $\int \eta\,\mathrm{d}\ln M = 0.1179$ and an episode of 5.3×10^7 yr, a factor 1.57 longer, which we carry as the conservative bracket. A figure of $\sim 10^8$ yr printed in earlier versions is withdrawn (Appendix C). The investment multiplies all subsequent per-unit-fuel returns by ∼5 (Fig. 3) and unlocks the BZ reservoir. The corollary for observers: spin is a key parameter of the system's history (Section 6), since the MTH requires engineered systems to approach near-extremal spin. This is the thin-disc spin-up branch (Paper E, Section 3.4; Swanson 2026a); a MAD-mode spin-up would instead end near $a_\star \approx 0.035$–0.07. The converse inference is weaker, because some natural formation channels, in particular coherent disk growth in a gas-rich nucleus, also reach high spin (Section 6; Berti and Volonteri, 2008; Reynolds, 2021).

4.2 Entropy disposal: the horizon as the ultimate cold reservoir

Landauer's principle sets the minimum thermodynamic cost of irreversible computation: erasing one bit dissipates at least $k_{\mathrm{B}}T\ln 2$ into a reservoir at temperature T (Landauer, 1961; Bennett, 1982; Parrondo et al., 2015), a bound verified experimentally (Bérut et al., 2012). Reversible logic evades the bound for intermediate steps (Bennett, 1973; Toffoli, 1980; Fredkin and Toffoli, 1982; Frank, 2002; Athas et al., 1994), but error correction, measurement, and memory reuse impose a residual erasure budget on any real computer (Wolpert, 2019). For a conventional deep-space computer the reservoir floor is the CMB at $T_\gamma = 2.7$ K, so the marginal cost is $k_{\mathrm{B}}T_\gamma \ln 2 = 2.6 \times 10^{-23}$ J per bit; the Landsberg-formalism treatment of radiative computation by Wright (2023) addresses this regime for Dyson-sphere computers, finding little advantage to nested shells and an optimum at small, hot spheres. This is the floor that aestivation proposes to lower by waiting $\sim 10^{12}$ years (Sandberg et al., 2016).

An event horizon lowers it in the present epoch. By the generalized second law (GSL), the sum of black-hole entropy $S_{\mathrm{BH}} = k_{\mathrm{B}}c^3A/4G\hbar$ and exterior entropy is non-decreasing, and entropy carried across the horizon is accounted by the area increase (Bekenstein, 1973, 1974; Bousso, 2002). A computation platform orbiting the hole may therefore dump its waste entropy into the horizon rather than radiating it to the sky. The marginal energy cost of horizon disposal is set by the horizon temperature

$$T_H = \frac{\hbar c^3}{8\pi G M k_{\mathrm{B}}} \simeq 6.2 \times 10^{-8} \left(\frac{M_\odot}{M}\right) \mathrm{K}, \tag{4}$$

(Hawking, 1974, 1975): $T_H \simeq 3 \times 10^{-12}$ K at $2 \times 10^4\,M_\odot$, twelve orders of magnitude below the CMB. A bit dumped at the horizon costs $k_{\mathrm{B}}T_H \ln 2 \sim 3 \times 10^{-35}$ J against 2.6×10^{-23} J for CMB radiation (Fig. 5). In practical terms the erasure cost ceases to be the binding constraint on computation; the budget shifts to transport of waste heat to the horizon (radiative or conductive tethering from the ISCO swarm inward), an engineering loss rather than a thermodynamic floor. The quantified gain from this substitution is the $\sim 10^{12}$ temperature ratio between the CMB and the horizon; the larger $\sim 10^{30}$ factor cited by Sandberg et al. (2016) assumes cooling to temperatures far below the horizon temperature used here and is not directly comparable. The substitution's decisive property is that it operates without waiting, and therefore in a setting where Bennett et al. (2019) showed that waiting for a colder background confers no advantage. The related configuration of Opatrný et al. (2017), a shell absorbing CMB radiation as its hot reservoir and rejecting entropy to a central black hole, demonstrates the same thermodynamic asymmetry in a fully worked classical setting.

The claim should also be stated in its energy-conservation form, because that is where the mainstream objection lands. Curtis et al. (2026) state the standard position: Dyson-scale computation must reradiate nearly all of the energy it absorbs, so mid-infrared thermal emission is the robust technosignature of black-hole-hosted intelligence. For a horizon-sink architecture the premise fails, and it fails on energy

grounds rather than entropy grounds: the waste carriers themselves cross the horizon, and their energy goes with the entropy. Writing $f_{\rm sink}$ for the fraction of waste *energy* delivered into the capture cone (a vocabulary adopted throughout this series), only the missed fraction $(1 - f_{\rm sink})$ must be reradiated, at swarm temperature; the absorbed remainder raises M and thereby lowers T_H further, so the channel is weakly self-improving. The disagreement with Curtis et al. (2026) is thus quantitative and falsifiable: the predicted mid-infrared residue scales with $(1 - f_{\rm sink})$ rather than with the full computational throughput.

The 2.7 K sky is also a competing cold sink, available without any black hole. Garrett (2026) argue that computation-dominated civilizations should radiate at the lowest temperature their surroundings permit, approaching the CMB floor; the required area grows as T^{-4}, the emission is grey, line-free, and submillimetre (a "Slysh halo"), and archival surveys already constrain such dissipation at the $\sim 10^{20}$ W level. The comparison of this section is measured against that alternative throughout: every gain quoted below stands against $k_{\rm B} T_\gamma \ln 2$, so the margin over a non-black-hole cold computer is of order unity (~ 0.05 on the conservative budget) if the carriers are thermally occupied by the 2.7 K sky, and $\sim 10^3$ in a fed flow or $\sim 10^7$ in the ambient medium only if carrier modes colder than the sky are kept from the emitter to the horizon, a delivery problem a cold radiator never faces. The case for the horizon sink over a cold radiator therefore rests on that condition: a sky-occluding aperture, and a path not thermalized by plasma at $T_e \gg 2.7$ K. Which architecture a maximizer prefers is not settled here; it turns on whether the transport budget of Table 2 is worth a sink twelve orders of magnitude deeper, and the two leave different residues, a submillimetre excess that scales with throughput against the $(1 - f_{\rm sink})$ mid-infrared remainder.

The $k_{\rm B} T_H \ln 2$ figure is the marginal cost of *disposal at the horizon*; it is not the delivered cost from an orbiting platform, and the gap between the two can be estimated rather than merely flagged. Consider radiative delivery, the least demanding channel. Efficient absorption requires a carrier wavelength $\lambda \lesssim r_g$, the geometric-optics condition for capture rather than diffraction around the hole (aiming is a separate and easier requirement, treated below). At $M = 2 \times 10^4\, M_\odot$, $r_g \simeq 3 \times 10^7$ m, so the vacuum-gravitational floor on the carrier energy is $E_\gamma \gtrsim hc/r_g \simeq 7 \times 10^{-33}$ J, corresponding to a frequency of $\sim$10 Hz. That floor is unreachable through plasma: propagation requires $\nu > \nu_p = 8.98\,{\rm kHz}\,\sqrt{n_e/{\rm cm}^{-3}}$, and even the cluster's ambient $n_e = 0.23\,{\rm cm}^{-3}$ (47 Tucanae analogy value; Abbate et al. 2018) gives $\nu_p \simeq 4.3$ kHz, while any fed inner-flow environment ($n_e \sim 10^4$–$10^8\,{\rm cm}^{-3}$ inside $\sim 10^2\, r_g$) gives $\nu_p \simeq$ 1–90 MHz. The minimum propagating carrier energy $E = h\nu_p$ therefore spans 2.9×10^{-30} J at the ambient density to 6.0×10^{-26} J at the densest fed flow, 2.6 to 6.9 decades above hc/r_g. Two consequences follow. First, single-bit-per-photon delivery no longer suffices: at the plasma-cutoff carrier energies the single-bit advantage over the CMB would erode to 9×10^6 in the ambient medium and to 4×10^2 in the densest fed flow considered. What preserves a realized gain is dense coding: a carrier may bear many bits, up to the channel capacity derived below, which is far under the Bekenstein bound (Bekenstein, 1981; Bousso, 2002). Coding at that capacity restores the delivered cost per bit to a factor of about 10^3 (fed flow) to 10^7 (ambient medium) below the CMB figure of 2.6×10^{-23} J, under the cold-mode condition stated below.

This requirement can be stated as a channel capacity rather than assumed. The Bekenstein bound caps the information carried by a packet of energy E confined to radius R at $I \le 2\pi E R/(\hbar c \ln 2)$ (Bekenstein, 1981), which for a one-wavelength packet gives at most 57 bits. Black holes saturate this bound; a photon carrier does not. For a pure-loss bosonic channel the capacity of one photon spread over M orthogonal modes is $\log_2(eM)$ bits for a lossless channel, and $\eta \log_2(eM/\eta)$ with transmissivity η (Giovannetti et al., 2004), so the achievable gain depends on the carrier floor and grows only logarithmically with packet length. On the conservative budget of Table 2, for a packet 100 km long, and for modes in the vacuum state, the realized gain over the 2.7 K sky is about 1.3×10^3 at the fed-flow floor (90 MHz) and about 9.3×10^6 at the ambient floor (4.3 kHz); the two floors differ by a factor of $\sim 7 \times 10^3$, because the mode count rises as λ^{-2} while the carrier energy falls. The Bekenstein figures remain in the table as ceilings.

Table 2: Delivery loss budget for the radiative channel and the realized gain it leaves for a packet 100 km long. The losses are the ones named in the text: error-correction overhead, capture and pointing efficiency into the 6.75×10^{-4} solid-angle target, transmission through the swarm without re-interception, and the ≈1 per cent lapse correction at $10^2\, r_g$. Throughput is their product. The first gain column is the Bekenstein ceiling, which a photon carrier does not reach; the last two are the gains from the pure-loss capacity of Appendix B at the fed-flow (90 MHz) and ambient (4.3 kHz) carrier floors, valid only for modes kept colder than the 2.7 K sky from emitter to horizon; with thermally occupied modes the conservative row falls to ∼ 0.05 at both floors.

Budget	Code rate	Capture f_{sink}	Swarm transm.	Lapse	Through-put	G ceiling (Bekenstein)	G fed (90 MHz)	G ambient (4.3 kHz)
Lossless (bound)	1.00	1.00	1.00	1.00	1.000	7.4×10^8	2.7×10^4	1.8×10^8
Optimistic	0.90	0.90	0.90	0.99	0.722	5.4×10^8	1.9×10^4	1.3×10^8
Conservative	0.50	0.30	0.30	0.99	0.045	3.3×10^7	1.3×10^3	9.3×10^6

The derivation is in Appendix B. The capacity formula holds when the noise source is in the vacuum state (Giovannetti et al., 2004). The 2.7 K sky has a mean occupation of about 630 photons per mode at 90 MHz and about 1.3×10^7 at 4.3 kHz, and carriers that share modes with it bear $\sim 2 \times 10^{-4}$ and $\sim 10^{-8}$ bits instead of tens of bits, which puts the conservative gain at ∼ 0.05 at both floors; by the Clausius inequality $dS \le dE/T_\gamma$ the disposal cost cannot fall far below the CMB figure for carriers thermalized at the sky temperature. The gains of $\sim 10^3$ and $\sim 10^7$ therefore hold only if the carriers travel in modes colder than 2.7 K from the emitter to the horizon, which needs a sky-occluding aperture and a path not thermalized by plasma at $T_e \gg 2.7$ K (Appendix B).

Delivery losses enter through the transmissivity and the code rate. Table 2 prices the losses the preceding paragraphs name and reports the gain that follows at 100 km of coherence. Under the cold-mode condition above, the conservative budget leaves about 10^3 in a fed flow and 10^7 in the ambient medium, and no budget or floor reaches 10^9; with thermally occupied modes it leaves ∼ 0.05. The Bekenstein column bounds the information a packet can hold and is no estimate of the rate.

The physical entropy of a gram of matter is 10^{22} to 10^{24} bits, far below the Bekenstein capacity of 2.6×10^{38} bits for a one-gram pellet at 100 km s^{-1} confined to 1 cm. The matter channel therefore delivers a gain of about 5×10^{-7} with kinetic charging and about 10^{-13} with opportunity-cost charging (ηmc^2 at $\eta = 0.30$), taking 10^{23} bits per gram, and does not compete with the 2.7 K sky.

Matter carriers evade the plasma cutoff, so cold mass dropped down the capture cone, which crosses the horizon with its entropy and energy aboard and no dispersion relation to satisfy, serves as a fallback for disposal where cheap photons are forbidden, although at a cost per bit above the CMB figure. With the loss budget of Table 2 charged against the radiative channel, the realized gain over the 2.7 K sky is about 10^3 in a fed flow and about 10^7 in the ambient medium if the carrier modes are kept colder than the sky (Section 4.2, Appendix B), and ∼ 0.05 if they are thermally occupied; the Bekenstein figure bounds the information a packet can hold and is no estimate of the rate. Gravitational bookkeeping does not change these figures: energy accounted at a platform of radius $r \gtrsim 10^2\, r_g$ is redshifted at infinity by the platform's lapse factor, a ≈1 per cent correction at these radii. The realized economy therefore sits well above the $k_{\text{B}} T_H \ln 2$ floor and well below the CMB figure, and we quote $\sim 10^3$ (fed flow) to $\sim 10^7$ (ambient), under the cold-mode condition, rather than the floor-to-floor ratio of $\sim 10^{12}$ wherever the realized gain is meant. An improvement of three orders of magnitude over the 2.7 K sky in a fed flow, available only under the cold-mode condition, is a large margin against any comparator that cannot reach below that sky, though smaller than the six orders quoted in earlier drafts.

Delivery geometry can be derived rather than assumed. The photon-capture cross-section of a Schwarzschild hole is $\sigma = \pi b_{\text{crit}}^2 = 27\pi r_g^2$, with critical impact parameter $b_{\text{crit}} = 3\sqrt{3}\, r_g$; from a platform

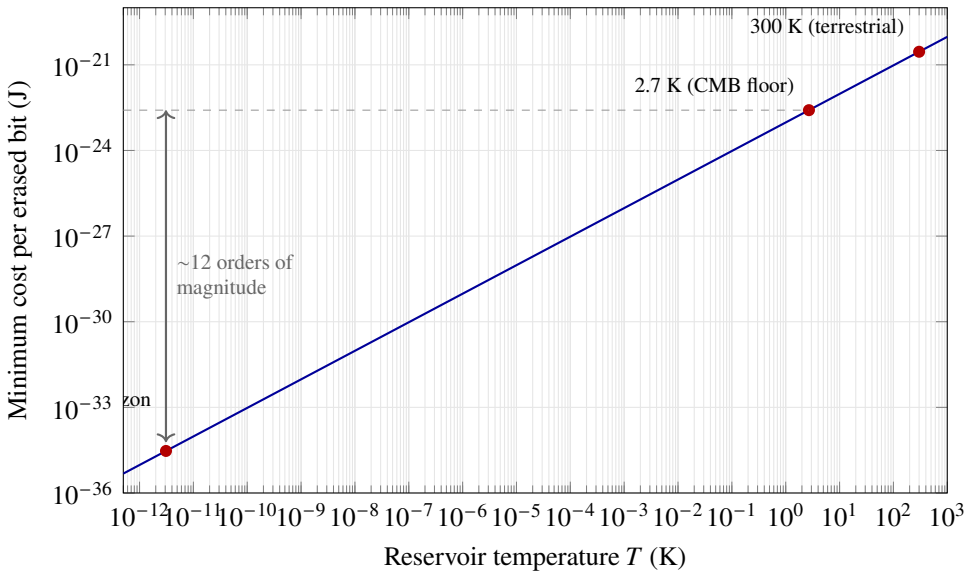


Figure 5: Landauer cost per erased bit, $k_B T \ln 2$, as a function of reservoir temperature. Dumping waste entropy across an IMBH horizon (marginal temperature T_H, Eq. 4) rather than radiating to the CMB lowers the erasure floor by $\sim 10^{12}$, in the present epoch, consistent with the generalized second law (Bekenstein, 1974). The aestivation hypothesis (Sandberg et al., 2016) sought a comparable-in-kind gain by waiting $\sim 10^{12}$ years for cosmological cooling, to still lower assumed reservoir temperatures.

at $r = 10^2\, r_g$ the capture target subtends a solid-angle fraction $\Omega_c/4\pi = 6.75 \times 10^{-4}$, so isotropic emission delivers less than a thousandth of its power to the horizon. Reaching $f_{\rm sink} \to 1$ therefore requires a beaming gain $\gtrsim 1.5 \times 10^3$, which is optically trivial: the capture cone has half-angle $\theta_c \simeq 0.052$ rad, so diffraction demands only an aperture $D \gtrsim 23\lambda$, and the étendue (concentration) limit $1/\sin^2\theta_c \approx 3.7 \times 10^2$ puts the optic area at a few hundred times the radiator area. (The $27\pi r_g^2$ figure is the Schwarzschild value; the near-extremal prograde equatorial cross-section is somewhat smaller and anisotropic, an $O(1)$ correction.) The binding term in the delivery budget is re-interception within the swarm rather than aiming or aperture: beamed flux must traverse the swarm itself, and the intercepted fraction scales with the swarm covering fraction (Swanson, 2026a).

4.3 Information storage: the Bekenstein–Hawking archive

The Bekenstein bound limits the information content of any bounded system (Bekenstein, 1981; Casini, 2008); black holes saturate it, with

$$\frac{S_{\rm BH}}{k_B \ln 2} \simeq 1.5 \times 10^{77} \left(\frac{M}{M_\odot}\right)^2 \text{bits}, \tag{5}$$

i.e. $\sim 6 \times 10^{85}$ bits at $2 \times 10^4\, M_\odot$ (Bekenstein, 1973; Hawking, 1975; Lloyd, 2000). That coefficient is the Schwarzschild value; since the architecture ends near-extremal, we quote one self-consistent pair: at $a_\star = 0.998$ the Kerr horizon area per unit M^2 is 0.53× Schwarzschild, so a $2 \times 10^4\, M_\odot$ hole at that spin stores $\approx 3 \times 10^{85}$ bits, and the factor-2.2 mass growth of the spin-up path (Section 4.1) more than recovers the difference through the M^2 scaling. Information crossing the horizon is scrambled but, by unitarity, not destroyed; retrieval timescales via Hawking radiation are, however, of order the evaporation time, $\sim 10^{67}(M/M_\odot)^3$ yr (Page, 1976): the archive is functionally write-only. Black holes likewise saturate the

Margolus–Levitin bound on processing rate, $2E/\pi\hbar$ operations per second (Margolus and Levitin, 1998; Lloyd, 2000). We treat these saturation properties as the formal statement that no material technology can out-store or out-compute the object the civilization is already orbiting.

4.4 Time dilation as an engineering parameter

A platform on a circular equatorial geodesic at radius r around a Kerr hole runs slow relative to infinity by

$$\frac{d\tau}{dt} = \frac{\sqrt{1 - 3r_g/r + 2a_\star(r_g/r)^{3/2}}}{1 + a_\star(r_g/r)^{3/2}}, \tag{6}$$

(Bardeen et al., 1972): $d\tau/dt = \sqrt{1/2} \simeq 0.707$ at the Schwarzschild ISCO, falling to $\simeq 0.093$ (a factor $\sim$11) at the prograde ISCO for $a_\star = 0.998$ (Fig. 6). Larger factors require powered non-geodesic hovering at diverging thrust cost and are not usable by stable platforms. The $\sim$11:1 ceiling is set by the Thorne (1974) spin-equilibrium limit $a_\star = 0.998$ rather than by orbital stability: the ISCO proper-time ratio falls toward zero as $a_\star \to 1$, and it is the accretion physics capping the spin that caps the dilation.[2] A tiered swarm can therefore place archival and slow-integration processes deep (subjectively skipping across external time) and interactive processes high, with the choice of orbit acting as a clock-rate dial, a resource with no terrestrial analogue, though we note its strategic value (e.g. outwaiting external change) trades directly against responsiveness. The dial is two-way: viewed from outside, a deep tier both runs and receives 11× slower in external time, so deep placement taxes bandwidth to, and coherence with, the shallow tier; and uplink photons arrive blueshifted by the same factor, each depositing $\sim$11× its emitted energy at the deep receiver.

4.5 Inward versus outward migration

The expansionist default implicitly maximizes resource acquisition; P1 civilizations maximize computation, and the two diverge. Expansion multiplies harvested mass polynomially in time ($\propto t^3$ for spherical growth at fixed speed) while imposing light-lag decoherence on any shared computation: a civilization spread over 10^3 light-years cannot function as one computer on subjective timescales shorter than millennia. Concentration, by contrast, multiplies the yield per unit mass by the factors of Section 4.1 (up to $\sim$600 over Dyson harvesting without star lifting at the Thorne cap), lowers the realized erasure cost by $\sim 10^3$ in a fed flow and $\sim 10^7$ in the ambient medium, if cold carrier modes are kept from the emitter to the horizon (Section 4.2), and keeps the entire system within light-seconds of itself. We must be careful about what this establishes. An undiscounted maximizer of total integrated computation over unbounded time is served by expansion despite the per-unit-mass penalty, because controlled mass grows as t^3 and per-unit efficiency gains are bounded. The gradient points inward only under at least one of the three conditions already carried in P1: (i) temporal discounting, so that near-term computation outweighs the polynomial tail; (ii) a requirement that the computation be unified and latency-bound, so that light-lag decoherence caps the useful radius; or (iii) bounded planning horizons, within which the concentration factors dominate the t^3 growth. Under any of these, beamed-sail transport makes a cluster-hosted IMBH reachable at negligible marginal cost (Lubin, 2016; Armstrong and Sandberg, 2013) and concentration

[2] Reiss and Feng (2026) have independently proposed time-dilated frames as a settlement strategy: biologically tolerable orbits near the photon radius of Sgr A* can, on their analysis, be maintained below the Kardashev Type II threshold, and a time-dilation factor of 10^4 enables galaxy-spanning migration within a subjective lifetime. They also note that isotropic, monochromatic signals emitted from such orbits exhibit a downward frequency drift. Their supermassive-host, near-photon-radius regime lies outside the stable-geodesic ceiling computed here.

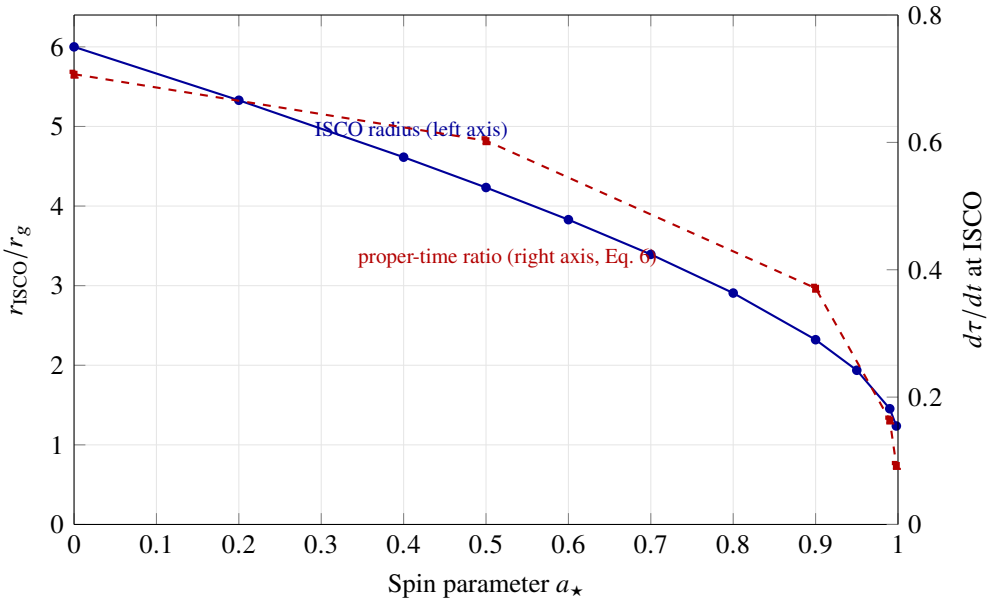


Figure 6: Prograde ISCO radius (left axis) and proper-time ratio for a circular geodesic at the ISCO (right axis) as functions of spin (Bardeen et al., 1972). Spin-up from $a_\star = 0$ to the Thorne limit moves the working orbit from $6\,r_g$ to $1.24\,r_g$ and deepens the available stable time dilation from 1.41:1 to ~11:1.

wins. We stress the modesty of the claim actually needed: P1 requires only that some civilizations satisfy one of (i)–(iii) and follow the gradient. Expansionist and stay-at-home civilizations may coexist with migrators; the Fermi-relevant point is that the oldest and most capable optimizers, the very civilizations whose absence the paradox finds most puzzling (Hanson et al., 2021), are the ones the gradient captures.

4.6 One hole rather than many

The comparison set for P2 must include a rival platform that the falsification program itself puts on the table: the dark remnant swarm of hypothesis H_1 (Section 6). Under H_1 the centre of ω Cen contains an extended $2\text{–}3 \times 10^5\,M_\odot$ component of stellar-mass black holes (Bañares-Hernández et al., 2025), twelve times the mass of the fiducial $2 \times 10^4\,M_\odot$ IMBH. Because $L_{\mathrm{Edd}} \propto M$, the swarm's aggregate Eddington power is $\sim 3.2 \times 10^{36}$ W against the IMBH's 2.5×10^{35} W: on raw power the "falsifying" configuration is the better platform by a factor of ~13, and a selection theory that ignored this would be assuming its conclusion. Table 3 prices the two configurations against the P2 axes.

The IMBH wins decisively on three of the four P2 axes and loses on the fourth. Storage scales as $\sum M_i^2$, so fragmenting the central mass into 2.5×10^4 holes of $10\,M_\odot$ costs a factor of 160 even against the twelvefold mass advantage. Sink depth scales as $T_H \propto M^{-1}$, a 2000× edge for the IMBH, and the edge survives the delivery-level accounting of Section 4.2: the realized erasure economy tracks the delivered carrier cost rather than T_H directly, and the vacuum carrier floor hc/r_g is 1.3×10^{-29} J for a $10\,M_\odot$ hole against 6.7×10^{-33} J for the IMBH, the same $\sim 2 \times 10^3$ ratio. The ISCO tidal field, $\propto M^{-2}$ at fixed r/r_g, is 4×10^6 times harsher at $10\,M_\odot$, excluding all but dust-scale hardware from the deep orbits where the dilation and efficiency gains live. And a computation distributed over the swarm pays parsec-scale light lag between nodes, against milliseconds across a single working orbit, failing the latency-bound condition of Section 4.5 by construction. A computation-maximizer offered both configurations takes the single hole; consolidation is worth a large premium under the same objective.

Table 3: One IMBH versus a same-order swarm of stellar-mass holes, priced on the P2 axes. The swarm column takes 2.5×10^4 holes of $10\,M_\odot$ (total $2.5 \times 10^5\,M_\odot$, the upper end of the Bañares-Hernández et al. 2025 extended component); the IMBH column takes the fiducial $2 \times 10^4\,M_\odot$. Storage and tidal entries are quoted as ratios (arb. units).

Figure of merit	One $2 \times 10^4\,M_\odot$ hole	2.5×10^4 holes of $10\,M_\odot$	Advantage
Aggregate $L_{\rm Edd}$	2.5×10^{35} W	3.2×10^{36} W	swarm, 13×
Storage, $\sum S_i \propto \sum M_i^2$	4×10^8	2.5×10^6	IMBH, 160×
Sink depth, $T_H \propto M^{-1}$	3.1×10^{-12} K	6.2×10^{-9} K	IMBH, 2000×
ISCO tidal field, $\propto M^{-2}$	1	4×10^6	IMBH, 4×10^6
Latency across working system	ms (one orbit)	μs per hole; pc-scale between holes	IMBH

The premium is a statement about preference, not about reach, and the gap between the two bounds what T3 can carry. Consolidating a remnant swarm means merging stellar-mass holes, and merger products receive gravitational-wave recoil kicks of 10^2–10^3 km s^{-1} for generic spin orientations (Campanelli et al., 2007; Lousto and Zlochower, 2011), against a central escape speed of some tens of km s^{-1} in ω Cen. Consolidation by merger therefore ejects most of its own products, and we know of no mechanism by which a civilization could assemble 2.5×10^4 holes into one while retaining them. The subsection establishes that a maximizer would prefer one hole. It does not establish that one could be built.

This derivation does real work in the falsification program. T3 (Table 7) treats a LISA-resolved remnant swarm as a decisive falsifier, and this subsection is why: the swarm loses on storage, sink depth, and tides by two to six orders of magnitude, so a maximizer with gigayears of run of the cluster would not have left the central mass in that configuration. T3 is thereby a derived kill switch rather than a stipulated one, conditional on the preference argument above transferring to a reachability claim, which the recoil problem leaves open. A resolved swarm is decisive against H_2 only if a maximizer with gigayears of run of the cluster could have consolidated it; if consolidation is dynamically blocked, a swarm is evidence about the cluster's formation history, not about whether anyone was there.

5 Omega Centauri as the optimal observational target

5.1 The cluster

ω Centauri is the most massive Galactic globular cluster ($M_{\rm cl} \approx 4 \times 10^6\,M_\odot$, $\sim 10^7$ stars; Harris 1996), at a kinematic distance of 5.49 ± 0.06 kpc (Häberle et al., 2025), in $\sim 2\sigma$ tension with the Gaia EDR3 parallax (5.24 ± 0.11 kpc; Soltis et al. 2021), whose error budget is dominated by the parallax zero-point systematics discussed by those authors, and consistent with the combined catalogue value (5.43 ± 0.05 kpc; Baumgardt and Vasiliev 2021); nothing in this paper depends on the difference. It is anomalous among globular clusters in nearly every respect that matters here: it hosts multiple stellar populations spanning a broad metallicity range (Johnson and Pilachowski, 2010), a mean stellar age of 12.08 ± 0.01 Gyr with an intrinsic spread of 0.75 Gyr (Clontz et al., 2024), significant rotation, and an associated tidal stream (Ibata et al., 2019); the consensus interpretation is that it is the surviving nucleus of a dwarf galaxy accreted and stripped by the Milky Way (Hilker and Richtler, 2000; Bekki and Tsujimoto, 2019; Souza et al., 2026). Table 4 summarizes the parameters used in this paper.

For the MTH selection criteria of Section 3.1, the stripped-nucleus origin matters twice over. First, nuclear star clusters are the environments most likely to form and retain IMBHs (Greene et al., 2020). Second, the age structure implies that any technological lineage arising anywhere in the progenitor dwarf

Table 4: Adopted parameters for ω Centauri (NGC 5139).

Quantity	Value	Source
Distance	5.49 ± 0.06 kpc	Häberle et al. (2025)
Cluster mass	$\approx 4 \times 10^6\ M_\odot$	Harris (1996); Baumgardt and Vasiliev (2021)
Stellar count	$\sim 10^7$	Harris (1996)
Mean stellar age	12.08 ± 0.01 Gyr (spread 0.75 Gyr)	Clontz et al. (2024)
Origin	stripped dwarf-galaxy nucleus	Hilker and Richtler (2000); Ibata et al. (2019); Souza et al. (2026)
Central density (catalogue)	$\sim 3 \times 10^3\ M_\odot\ \mathrm{pc}^{-3}$	Baumgardt and Vasiliev (2021); cf. $\sim 10^3$ in Pryor and Meylan (1993)
Central density (King model, adopted)	$4.0 \times 10^3\ M_\odot\ \mathrm{pc}^{-3}$	Section 5.2; used for every enclosed-mass figure
Declination	$-47.5°$	Harris (1996)

(or in the early Milky Way) has had $\gtrsim$8 Gyr to locate, reach, and develop the system: at beamed-sail transit speeds of 0.1–0.2 c (Lubin, 2016), crossing the Galaxy takes $\lesssim 10^6$ years, roughly eight thousand times shorter than the available window.

See Paper F App. A for the shared-parameter table across the set, which cross-references this table's distance, cluster mass, and central-density rows against the other papers' own choices.

5.2 The enclosed-mass profile

The fuel-census figures used in this paper (Sections 3.1 and A) are referred throughout to one stellar enclosed-mass profile, stated here so that every ratio is reconstructible. We integrate a single-mass lowered-isothermal (King 1962) model to the catalogued concentration $c = \log_{10}(r_t/r_c) = 1.31$ and core radius $r_c = 2.37'$ (Harris, 1996), scaled at our adopted distance (Table 4), and normalize to the total cluster mass $4 \times 10^6\ M_\odot$. This gives $W_0 = 6.22$, $r_t = 77$ pc, and a central density $\rho_c = 4.03 \times 10^3\ M_\odot\ \mathrm{pc}^{-3}$. That is 34 per cent above the catalogued $\sim 3 \times 10^3$ of Table 4, and the two are not interchangeable: a flat-core estimate of $M(< 1\ \mathrm{pc})$ lands within 6 per cent of the integrated profile at 4.0×10^3 and 20 per cent low at 3×10^3. Every enclosed-mass, fuel-census and depletion figure in this paper is referred to the model value, and we state it once here rather than quoting the two as though they agreed; the catalogued value is retained in Table 4 as the external anchor the model is scaled against through c and r_c, not through ρ_c. The model's projected half-mass radius, 8.3 pc (5.2′), matches the catalogued 5.00′ to 3–4 per cent. Enclosed stellar mass:

The profile is a stellar background and carries no mass segregation; a multi-mass treatment is not needed for any figure quoted in this paper.

5.3 The IMBH candidate and the mass tension

Evidence for a central dark mass in ω Cen has accumulated for two decades, from integrated kinematics (Noyola et al., 2008) through proper-motion modelling that initially yielded only upper limits at the $\sim 10^4\ M_\odot$ level (van der Marel and Anderson, 2010), with stellar-mass black-hole subsystems long advanced as the alternative explanation (Zocchi et al., 2019; Breen and Heggie, 2013) and a systematic N-body survey concluding that neither ω Cen nor NGC 6624 required an IMBH (Baumgardt et al., 2019). The situation changed qualitatively when Häberle et al. (2024b), using a proper-motion catalogue of 1.4 million

Table 5: Stellar enclosed mass from the King profile of Section 5.2. Star counts convert at 0.43 $M_\odot$ per star. The two duration columns answer different questions. Column 4 expresses the reservoir as Eddington-years at a hole held fixed at 2×10^4 $M_\odot$ (2,250 yr per solar mass, Section 4.1); it is linear in fuel mass and states the reservoir size; it is not an elapsed time. Column 5 is the elapsed time to actually consume the reservoir, $t_S \ln(1 + M_{\rm fuel}/M_i)$ with $t_S = \eta c^2/(L_{\rm Edd}/M) = 4.5 \times 10^7$ yr at $\eta = 0.1$, which accounts for the Eddington limit rising as the hole grows. The 2-pc and 10-pc rows are bookkeeping rather than scenarios: their endpoint masses, 1.3×10^5 and 1.9×10^6 $M_\odot$, leave the 10^3–10^5 $M_\odot$ range of selection criterion 1, so a hole that consumed either reservoir would no longer be the object this paper selects.

r	$M(<r)$ ($M_\odot$)	stars	Fuel at fixed M (Eddington-yr)	Elapsed (growth-aware)	Endpoint M ($M_\odot$)
1 pc	1.6×10^4	3.7×10^4	3.6×10^7	2.6×10^7 yr	3.6×10^4
2 pc	1.1×10^5	2.5×10^5	2.4×10^8	8.3×10^7 yr	1.3×10^5
10 pc	1.9×10^6	4.3×10^6	4.2×10^9	2.0×10^8 yr	1.9×10^6

stars built from over 500 HST epochs (Häberle et al., 2024a), identified seven stars within 3 arcsec of the cluster centre moving faster than the local escape velocity, of which five are robust detections. Their velocities on those five stars alone require a point mass $\geq$ 8,200 $M_\odot$; including acceleration limits on the same stars raises the lower bound to 21,100 $M_\odot$ at 99 per cent confidence (main text of Häberle et al. 2024b).[3] Independent N-body modelling finds that an IMBH grown in situ to $\sim 5 \times 10^4$ $M_\odot$ reproduces the present-day cluster structure and fast-star population (González Prieto et al., 2025).

Against this stands Bañares-Hernández et al. (2025): a joint analysis of stellar kinematics and timing accelerations of the cluster's millisecond pulsars that places a 3σ upper limit of $\sim 6 \times 10^3$ $M_\odot$ on any central point mass, favouring instead an extended central dark component of $\approx$ 2–3 $\times 10^5$ $M_\odot$, naturally interpreted as a centrally concentrated cluster of stellar remnants. The most recent joint MeerKAT–Parkes timing analysis, by contrast, finds its measurements insensitive to an IMBH in the 10^3–10^4 $M_\odot$ range and places a 90 per cent upper limit of $< 10^5$ $M_\odot$ on the central mass (Colom i Bernadich et al., 2026), a bound compatible with both camps. The two leading results are formally inconsistent under each side's stated modelling assumptions, and the disagreement is unresolved at the time of writing (Table 6). We take no side; for this paper, the tension is what makes the system valuable as a target. The MTH requires a genuine IMBH (P2), so the Bañares-Hernández et al. (2025) scenario, if confirmed, falsifies the ω Cen application outright (one of several clean kill switches catalogued in Section 6). LISA can settle the question definitively, contingent on a suitable source being in-band during the mission: a single IMBH undergoing a compact-object inspiral yields resolvable extreme- or intermediate-mass-ratio-inspiral (EMRI/IMRI) signals with percent-level mass and spin measurement, whereas a remnant swarm yields a qualitatively different gravitational-wave signature (Amaro-Seoane, 2018; Babak et al., 2017; Colpi et al., 2024).

The contingency deserves emphasis, because the per-cluster IMRI rate for a $\sim 10^4$ $M_\odot$ hole is uncertain by orders of magnitude, and the plausible outcome of a 4–10 yr mission is that no inspiral is caught in-band, in which case the gravitational-wave branch of Section 6 returns "inconclusive" rather than a verdict (Fig. 7, silence branch). Two mitigations exist. The dense fast-star environment resolved by Häberle et al. (2024b) raises the prior on close compact companions relative to a field IMBH, though it does not remove the rate uncertainty. More important, an inspiral is not the only LISA channel: Strokov et al. (2022) show that a Galactic-cluster IMBH can reveal itself through Doppler modulation of the gravitational-wave

[3]The abstract of Häberle et al. (2024b) quotes only the velocity-only 8,200 $M_\odot$ limit; the 21,100 $M_\odot$ figure appears in the main text. A Nature *Author Correction* (doi:10.1038/s41586-024-08017-4) fixes an equation typo and a right-ascension offset; the results are unaffected.

Table 6: Current constraints on the central dark mass of ω Cen. The kinematic lower bounds and the pulsar-timing upper bound are mutually inconsistent under stated assumptions; resolution is expected from LISA EMRI observations and continued astrometry.

Constraint	Value	Method / source
Lower bound (velocities only)	$\geq 8{,}200\,M_\odot$	5 of 7 fast stars (robust subset), HST proper motions (Häberle et al., 2024b)
Lower bound (acceleration-model-informed)	$\geq 21{,}100\,M_\odot$ (99%)	same stars, acceleration limits (Häberle et al., 2024b)
N-body growth models	~ 4.7–$5.1 \times 10^4\,M_\odot$	cluster-evolution simulations (González Prieto et al., 2025)
Upper bound (point mass)	$< 6 \times 10^3\,M_\odot$ (3σ)	stellar kinematics + MSP timing (Bañares-Hernández et al., 2025)
Favoured alternative	extended 2–$3 \times 10^5\,M_\odot$	same analysis (Bañares-Hernández et al., 2025)
Timing (joint MeerKAT–Parkes)	insensitive at 10^3–$10^4\,M_\odot$; $< 10^5\,M_\odot$ (90%)	MSP timing (Colom i Bernadich et al., 2026)
Historical upper limits	$\lesssim 10^4\,M_\odot$ (model-dep.)	HST proper motions (van der Marel and Anderson, 2010); see also Zocchi et al. (2019)

signal of an orbiting stellar-mass binary, a continuous signature that requires no merger event during the mission. This channel constrains the mass, though not the spin, so the decisive T5 measurement remains inspiral-contingent.

One element of the predicted dark population has now been observed directly. Whitaker et al. (2026) report the astrometric detection of a $4.46^{+1.22}_{-1.01}\,M_\odot$ black hole with reported orbital elements $P = 94$ yr, $e = 0.72$, $a = 31$ AU and a $\sim 0.8\,M_\odot$ main-sequence companion,[4] from HST astrometry spanning 2002–2023 combined with JWST imaging: the first confirmed stellar-mass black hole in ω Cen, and the first astrometric detection of one in any globular cluster. The detection confirms that the cluster forms and retains stellar-mass black holes, the population of which both sides of the mass tension require ($\sim 10^4$ such remnants in the extended-component scenario; a formation reservoir in the IMBH-growth scenario), and it anchors the low-mass end of the central mass budget with an object whose parameters are measured rather than modelled.

5.4 Electromagnetic silence

The centre of ω Cen is extraordinarily dark, whatever occupies it. The deepest radio observation of any globular cluster ($\sim$170 h with ATCA at 7.25 GHz, reaching an rms of 1.1 μJy beam^{-1}) detects nothing at any proposed cluster centre, bounding a putative IMBH accreting at the Bondi rate to a model-specific accretion fraction $\lesssim 4 \times 10^{-3}$ (3σ; model-dependent, with $\approx$1 dex of systematic from the fundamental-plane conversion) under Mahida et al.'s own radiative/jet assumptions (Mahida et al., 2026), not the accretion efficiency $\epsilon \equiv L_{\rm bol}/(\dot{M}_{\rm Bondi} c^2)$ (see Paper F's convention table, Swanson 2026b), the deepest point yet in the radio-limit framework established for globular clusters by the MAVERIC survey

[4] We note without resolution that the reported elements are in mild internal tension: Kepler's third law with $a = 31$ AU and $M_{\rm tot} \simeq 5.3\,M_\odot$ gives $P \simeq 75$ yr; the reported period exceeds this Kepler-implied value by $\sim$25 per cent; reconciliation lies within the quoted uncertainties, since the $^{+1.22}_{-1.01}\,M_\odot$ range on the black-hole mass alone moves $M_{\rm tot}$ between 4.3 and 6.5 $M_\odot$ and hence the Kepler-implied period between 83 and 68 yr at fixed a, while the remaining residual, $94/83 \simeq 1.13$ at the longest Kepler-implied period (83 yr, $M_{\rm tot} = 4.3\,M_\odot$), closes with a semi-major axis larger by $\sim$8 per cent, since $P \propto a^{3/2}$ ($1.13^{2/3} \simeq 1.08$).

(Tremou et al., 2018). JWST NIRCam and MIRI photometry of the central field likewise finds no source with the spectral energy distribution of an accreting IMBH, with limits most constraining at the low-mass end of the allowed range (Chen et al., 2026). The X-ray channel, the standard probe of quiescent accretion, agrees: a ∼291 ks Chandra exposure detects no central point source, limiting the unabsorbed luminosity to $L_X(0.5\text{–}7.0\,\mathrm{keV}) \lesssim 1.6 \times 10^{30}\,\mathrm{erg\,s^{-1}}$, an Eddington ratio of 1.3×10^{-12} for a $10^4\,M_\odot$ hole, or 6×10^{-13} at the $2 \times 10^4\,M_\odot$ fiducial (Haggard et al., 2013). To our knowledge, no dedicated technosignature search of ω Cen has been conducted at any wavelength, although the commensal opportunity is unusually rich: ω Cen is a flagship MeerKAT pulsar-timing target, so every timing session is a potential commensal SETI epoch on the cluster for Breakthrough Listen's MeerKAT backend; the first dedicated globular-cluster SETI survey (Huang et al., 2026b, five northern clusters with FAST; ω Cen is inaccessible from FAST's latitude) demonstrates both the feasibility and the current emptiness of this niche, and a Gaia-based phase-space anomaly metric over 79 Galactic globular clusters, in which NGC 5139 ranks among the dynamically most complex systems proposed for technosignature-oriented follow-up (Huang et al., 2026a), provides independent validation for treating globular-cluster anomalies as search targets.

The logic matters here: the MTH predicts electromagnetic silence (P3–P4), but silence is equally consistent with, and more parsimoniously explained by, a quiescent, gas-starved compact object in a relaxed, gas-poor old cluster, which we adopt as the null hypothesis H_0 throughout. Silence cannot confirm the MTH; only the discriminating residues of P4 can separate the hypotheses (Section 6).

5.5 The case against Sagittarius A*

A natural objection: if bigger is better, the Galactic Centre hosts a black hole of mass $4.3 \times 10^6\,M_\odot$ (GRAVITY Collaboration, 2022), imaged directly as a shadow (Event Horizon Telescope Collaboration, 2022), with abundant fuel. Three of the four selection criteria of Section 3.1 disfavour it; the first is thermodynamic, the second and third strategic (per the register of claim 1, Section 1.1). First, the storage and erasure advantages saturate at IMBH scale: nothing a maximizer needs scales usefully beyond $\sim 10^{4\text{–}5}\,M_\odot$, while hazards do. Second, the Galactic Centre is the most dynamically violent environment in the Galaxy (ongoing star formation, supernovae, magnetar flares, stochastic accretion flares, and a dense population of perturbers), the opposite of a stable substrate for Gyr-scale infrastructure; ω Cen's relaxed core offers the same physics in a far quieter dynamical environment. Third, the claim structure matters, in the era-relative form of criterion 3: the Galactic Centre's gas and stars are the future fuel of natural astrophysics and of any fusion-era competitor, whereas a globular cluster's low-mass reservoir is uncontested by fusion-era economics and contested only among computation-maximizing lineages, the population P1 already conditions on; the premium therefore goes to early arrival rather than to contest avoidance. The MTH thus predicts that the preferred class is IMBHs in dense old clusters rather than supermassive holes, a distinctive and testable preference, since it directs attention at objects mainstream SETI has never examined.

6 Falsification framework and observational program

We define the competing hypotheses explicitly, then the tests.

H_0 **(gas starvation; null).** The central object, whether IMBH or remnant swarm, is electromagnetically silent for the ordinary reason: a relaxed, gas-poor, 12-Gyr-old cluster supplies nothing to accrete. Favoured by parsimony and adopted as default.

H_1 **(dark remnant swarm).** No IMBH exists; the central mass is an extended cluster of stellar remnants (Bañares-Hernández et al., 2025; Zocchi et al., 2019; Breen and Heggie, 2013). Falsifies the ω Cen application of the MTH.

H_2 **(MTH).** An IMBH exists and is, or has been, the substrate of a migrated civilization (Appendix A).

Four load-bearing numbers below are taken from the companion infrastructure analysis (Swanson, 2026a): the T2 waste-heat floor $L_{\rm waste} \gtrsim 10^{-4} P_{\rm comp}$, the Phase-3 erosion boundary, the $\sim 4 \times 10^{11}$ yr Blandford–Znajek spindown time behind the T5 natal-spin argument, and the $7.6 \times 10^5\, L_\odot$ ambient Bondi ceiling used in Appendix A. That companion is a preprint and is not yet refereed. We mark the thresholds that depend on it provisional, and a reader who declines the companion should treat T2 and T5 as stated without numerical thresholds.

H_0 and H_2 predict identical electromagnetic appearances today (by design, since P3–P4 entail silence), so electromagnetic non-detections cannot distinguish them. They diverge on dynamical and high-energy observables. Table 7 states six tests with thresholds, instruments, and timelines; Fig. 7 arranges the decisive subset as a decision tree.

The matrix divides into two kinds of test, and the division matters. T1, T2 and T6 are constraint channels: since H_0 and H_2 predict identical electromagnetic appearances, a deeper null in any of them lowers a ceiling on managed throughput and retires no phase, and there is no null threshold at which they adjudicate between the hypotheses. We state the consequence rather than leave it implicit: no accumulation of electromagnetic non-detections, however deep, can falsify the MTH. T3 and T5 are the decisive tests, T4 a ranking demotion, and all three are contingent on a compact-object inspiral occurring in LISA band during the mission. The MTH cannot survive a confirmed remnant swarm or a non-spinning IMBH (on the thin-disc branch, Paper E Section 3.4), except as vacuous speculation about other systems; but whether it is adjudicated at all in this generation depends on an event rate the paper cannot pin down (Section 5.3). Second, the LISA spin measurement is among the most informative numbers obtainable this generation (if an in-band inspiral delivers it; Section 5.3), though both the natural-spin null and the expected yield must be stated carefully. Two natural channels are on the table for ω Cen; adopting both at once would leave the high-spin branch with no discriminating power in either direction, so we adopt one. The paper's own account of ω Cen's origin (Section 5) is a gas-rich, accreted dwarf-galaxy nucleus, so the prior we carry is coherent disk growth, which generically reaches high spin (Berti and Volonteri, 2008; Reynolds, 2021); because Blandford–Znajek spindown at ambient fueling operates on $\sim 4 \times 10^{11}$ yr timescales (Swanson, 2026a), such natal spin survives to the present. Under that prior a measured $a_\star \gtrsim 0.9$ is expected on H_0 as well as on H_2 and is not by itself a residue. The alternative channels sharpen T5 rather than blunt it. Runaway stellar collisions build IMBHs of several $10^3\, M_\odot$ (Fujii et al., 2024), though those simulations do not report a remnant spin, so the channel leaves $a_\star$ unconstrained, not low. Growth by randomly oriented accretion episodes drives the spin to an equilibrium near $a_\star \approx 0.1$–0.3 with fluctuations of ± 0.2 (King et al., 2008; Berti and Volonteri, 2008), which places substantial prior weight at and below the T5 trigger $a_\star < 0.1$. What we cannot supply is the relative weight of the coherent and chaotic channels for this object, and the quantity the test turns on is therefore that weight, which we cannot supply. A measured $a_\star \gtrsim 0.9$ counts as supporting evidence only in conjunction: the discriminating observation is the conjunction of high spin, electromagnetic silence far below the ambient Bondi supply, and the regulation signature of T1 (P4a–b together). EMRI parameter estimation at LISA delivers spin to better than percent precision for favourable systems (Babak et al., 2017; Amaro-Seoane, 2018). Third, the program is cheap: T1–T2 are archival or piggyback analyses; T6 requires only a pointed monitoring program on an instrument already built for the southern sky (Adrián-Martínez et al., 2016; The KM3NeT Collaboration, 2025), following the high-energy SETI logic of Lacki and DiKerby (2025) and the pipeline practices of standard

Table 7: Falsification matrix for the MTH at ω Cen. "Kills" indicates what a positive result rules out; tests marked † are decisive against the hypothesis as a whole rather than against one phase. LISA timeline assumes launch in the mid-2030s (Colpi et al., 2024); T3–T5 are additionally contingent on a compact-object inspiral being in-band during the mission (Section 5.3), with the Doppler-modulation channel of Strokov et al. (2022) providing a mass-only fallback. T5 holds on the thin-disc spin-up branch (Paper E, Section 3.4); on a MAD-mode spin-up the endpoint is $a_\star \approx 0.035$–0.07, within 1σ of Paper E's natural-capture null and below the T5 trigger $a_\star < 0.1$, so the trigger could fire on an engineered hole.

#	Observation	Threshold	Instrument	Timeline	Kills
T1	Accretion luminosity from core (limit-tightening)	Current best mid-infrared bolometric-equivalent limit $L_{\rm acc} \approx 2.5 \times 10^{26}$ W ($\approx 10^{-9} L_{\rm Edd}$) (Chen et al., 2026), versus the deeper band-limited Chandra X-ray bound L_X(0.5–7.0 keV) $\lesssim 1.6 \times 10^{23}$ W ($\lesssim 10^{-12} L_{\rm Edd}$) (Haggard et al., 2013); near-term target one further decade, $\lesssim 2.5 \times 10^{25}$ W. Supplementary trigger: sustained (> 1 yr) brightening to $L_{\rm acc} > 10^{29}$ W	JWST MIRI; ATCA/ MeerKAT; Chandra	now–2030	Constraint channel, not a kill: each deepening null lowers the ceiling on managed accretion throughput without retiring any phase. A brightening detection kills Phase 5 (managed environment). Bounds the accretion SED itself, where T2 bounds reradiated waste heat (Haggard et al., 2013; Mahida et al., 2026)
T2	Mid-IR waste-heat excess	$> 3\sigma$ over stellar-population model, 10–25.5 μm (MIRI imaging limit); tested against the horizon-sink floor $L_{\rm waste} \gtrsim 10^{-4} P_{\rm comp}$ (Swanson, 2026a)	JWST MIRI	now–2030	Constraint channel, not a kill: any Dyson-type capture is excluded, and a deepening null tightens the ceiling on Phase 4–5 throughput (currently $P_{\rm comp} \lesssim 10^{3-4}\, L_\odot$). Because the horizon-sink floor is $L_{\rm waste} \gtrsim 10^{-4} P_{\rm comp}$, no null retires managed operation outright; it drives $P_{\rm comp}$ down without bound
T3†	GW signature inconsistent with single point mass	stochastic background or multiple low-mass inspirals from core	LISA	late 2030s–2040s	H_2 entirely (confirms H_1); removes P2 substrate
T4	EMRI point mass below the kinematic floor (demotion, not kill)	clean chirp, $M < 8.2 \times 10^3\, M_\odot$ at $> 3\sigma$; the trigger is the Häberle et al. (2024b) kinematic lower bound, not the $10^3\, M_\odot$ class edge	LISA	late 2030s–2040s	Demotes the selection ranking of Section 3.1: ω Cen would no longer be a P2-optimal destination, though an $8 \times 10^3\, M_\odot$ hole still delivers $\sim 1.0 \times 10^{35}$ W (Eq. 2), so H_2 is weakened rather than excluded
T5†	EMRI spin measurement low	$a_\star < 0.1$ at $> 3\sigma$	LISA	late 2030s–2040s	The exploitation model: no BZ reservoir, no spin-up history; MTH survives only in unmodified "pre-arrival" form; thin-disc branch only (note)
T6	High-energy neutrino bursts from core	≥ 3 tracks within 1° in 10^{2-3} s, $E \gtrsim 10$ TeV, $> 5\sigma$ post-trials, no astrophysical counterpart	KM3NeT/ARCA (ω Cen is up-going from the Mediterranean; dec. −47°)	2026–2035	Constraint channel, not a kill: a null at full sensitivity closes the Dvali and Osmanov (2023) channel only, which the MTH does not require; a detection is consistent with H_2 without proving it

point-source searches (Aartsen et al., 2020); the partially built ARCA array has already demonstrated discovery-class capability with the ∼220-PeV event KM3-230213A, the most energetic neutrino yet observed (The KM3NeT Collaboration, 2025); and T3–T5 are free by-products of LISA's planned EMRI science (Colpi et al., 2024).

The T6 burst criterion must clear the atmospheric-neutrino background, and an order-of-magnitude estimate shows it does under the stated cuts. Full-detector ARCA is expected to collect of order 10^3 up-going atmospheric muon-neutrino events per year above 10 TeV over the full up-going sky (effective areas and atmospheric fluxes from the KM3NeT 2.0 letter of intent; Adrián-Martínez et al., 2016). A 1°-radius cone subtends $\sim 10^{-3}$ sr, a fraction $\sim 1.5 \times 10^{-4}$ of the up-going hemisphere, giving a source-cone rate of ∼0.15 events yr^{-1}, i.e. $\lambda \sim 5 \times 10^{-9}$ s^{-1}. The Poisson expectation in a 10^3 s window is then $\mu \sim 5 \times 10^{-6}$, so the probability of an accidental ≥3-fold multiplet per window is $\mu^3/6 \sim 2 \times 10^{-17}$; a decade of livetime contains $\sim 3.2 \times 10^5$ independent 10^3 s windows, for an expected false multiplet count of $\sim 6.6 \times 10^{-12}$ per decade. The 1° cone is itself deliberately conservative: ARCA's track angular resolution above 10 TeV is ∼0.1–0.2°, so a resolution-matched cone carries a background lower by a further factor of 25–100. The criterion is background-free by many orders of magnitude even allowing generous uncertainties in the effective area and a trials factor for scanning window durations; the practical limit on T6 is exposure, not background. A dedicated ω Cen technosignature campaign would be the first of its kind for any globular cluster beyond the pilot survey of Huang et al. (2026b), which could not observe ω Cen.

6.1 Supporting evidence criteria

Symmetry requires stating the confirmation side. The MTH gains support only from conjunctions that H_0 finds awkward: (i) a LISA-confirmed IMBH with $a_\star \gtrsim 0.9$ in conjunction with Gyr-scale gas starvation and the regulation signature of T1, since high spin alone admits the natal-disk explanation above; (ii) repeated $> 5\sigma$ neutrino bursts from the core with hard spectra and no electromagnetic counterpart, matching the Dvali and Osmanov (2023) phenomenology; or (iii) a measured low-mass mass-function slope in the central 0.1 pc lying below the González Prieto et al. (2025) model envelope at $> 3\sigma$ (oMEGACat photometry reaches the relevant depth), the concrete form of the P4c depletion residue. Criterion (iii) is the weakest of the three and we mark it provisional. A 1.3 per cent removal from the 10-pc reservoir preserves the total enclosed mass and is only claimed to be observable because it alters the stellar mass function rather than the density profile; whether a removal of that size imprints a $> 3\sigma$ slope anomaly that survives ∼12 Gyr of mixing has not been forward-modelled, and until an N-body calculation supplies the predicted amplitude, (iii) states where to look without stating what to expect. No single line, including all three jointly, would constitute proof of technology; they would constitute an anomaly stack justifying escalated scrutiny. Conversely, we commit in advance: if LISA delivers T3, T4, or T5, the ω Cen application of the MTH should be retired without special pleading.

7 Discussion: objections and limits

Parsimony. The MTH posits extraterrestrial intelligence; H_0 posits nothing. By any reasonable prior the null is favoured, and we have structured every comparison accordingly. The defensible claim is that the MTH is cheap to test relative to its information value, not that it is probable: it concentrates the diffuse question "where is everybody?" onto a handful of named objects and a decade-scale instrument schedule, and its decisive observables (IMBH reality, mass, spin) are independently first-rank astrophysics that will be measured anyway.

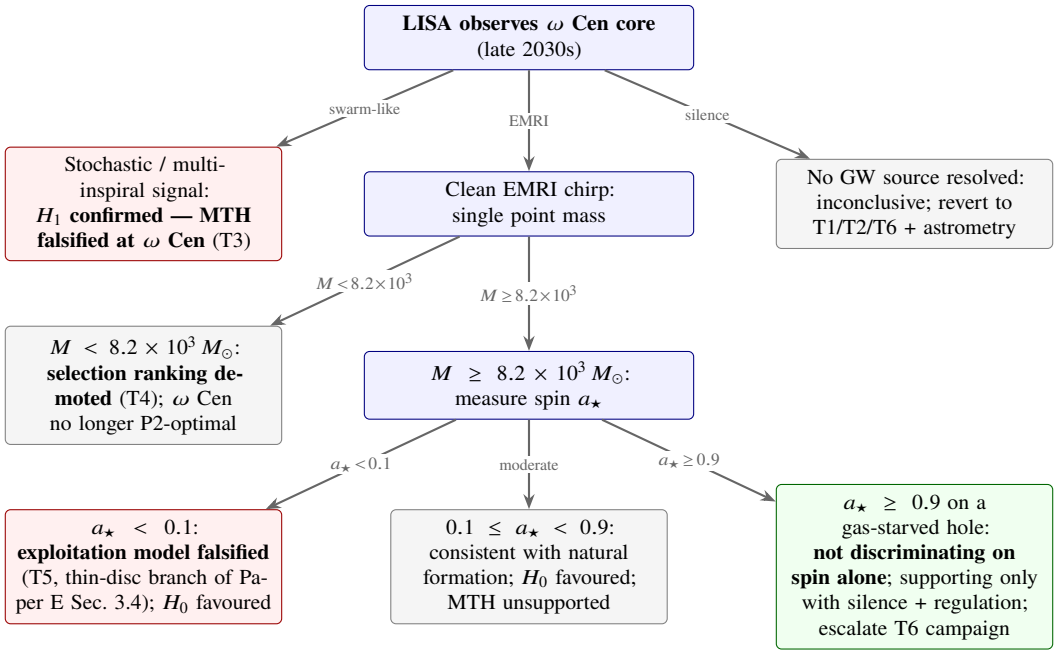


Figure 7: Decision tree for the decisive (gravitational-wave) branch of the falsification program. Red boxes terminate the hypothesis or its exploitation model; the single green branch is the only outcome in which the MTH gains material support, and even there the inference is to "anomaly requiring explanation," not detection. The mass branches are exhaustive at the $8.2 \times 10^3\ M_\odot$ kinematic lower bound of Häberle et al. (2024b), which is the threshold T4 uses; the earlier split at 5×10^3 and 10^4 left measured masses in $[5 \times 10^3, 10^4)$, the interval containing the kinematic bound itself, with no terminal (Appendix C). The $a_\star < 0.1$ node assumes the thin-disc spin-up branch of Paper E Section 3.4; on a MAD-mode spin-up it could fire on an engineered hole. Under the coherent-growth prior adopted in the text the high-spin branch is expected on H_0 too, which is why it is labelled as non-discriminating on spin alone.

The anthropic objection. If P1–P2 captured all civilizations, our own existence as a young, loud, planet-bound species is unremarkable; the MTH explains the silence of the old, not the existence of the young. It is therefore not undermined by our own counterexample, but neither can it explain why no expansionist civilizations are visible (cf. Hanson et al. 2021): if migration is optional, loud lineages should still exist somewhere. The defensible position is that the MTH thins the expected population of loud civilizations by removing its most capable members, sharpening rather than fully dissolving the paradox; full dissolution requires either high migration compliance or supplementary rarity from conventional Drake-equation factors.

Goal stability over 10^8 years. Phase 4 presupposes coherent purpose across timescales that dwarf all institutional experience. We flagged this as the architecture's largest unargued premise (Appendix A). One partial mitigation: the architecture's payoff structure is front-loaded (Phase 3 already yields a substrate orders of magnitude beyond planetary computing), so a lineage that fragments mid-program still leaves durable traces, but fewer than we once supposed. The companion analysis (Swanson, 2026a) finds that an abandoned swarm is not preserved: collisional cascade grinds it to debris within $10^{2–3}$ yr, and the debris drains into the hole. The durable residues of incoherence are the spin acquired before fragmentation and, if Phase 4 ran long enough, the core depletion it caused; the hardware itself leaves no fossil. The observational program is robust to civilizational incoherence only through those two channels, which is a further reason the dynamical tests (T3–T5) carry the weight.

Tidal and radiative survivability at the ISCO. At $2 \times 10^4\ M_\odot$ the tidal field at the Schwarzschild ISCO is $2GM/r_{\rm ISCO}^3 \simeq 1\ \mathrm{s}^{-2}$: a 10-m node experiences a benign $\sim 1\ g$ differential, while kilometre-scale rigid structures would face $\sim 10^2\ g$ — one reason the architecture assumes a swarm of small free-flying nodes rather than a monolithic platform. Radiation from even a managed accretion disk is the harsher constraint, and motivates the tiered-orbit design and sub-Eddington feeding discipline assumed in Appendix A. We have not engineered these systems in detail and do not pretend to; the claim defended is the thermodynamic gradient (P2), not any specific machine.

Why has nothing arrived here? A migration hypothesis must explain why migrators' probes are not conspicuous in every system, reviving Hart (1975) and Tipler (1980). The MTH answer is economic: for a computation-maximizer, planetary systems are at most transit infrastructure rather than destinations; the Galaxy's uncollected starlight (an integrated stellar luminosity of $(1–2) \times 10^{37}$ W; cf. Flynn et al. 2006) dwarfs any single system's output by ten orders of magnitude, and yet (Section 4.5) the marginal value of any outward claim is dominated by deepening the home gravity well. Quiet, minimal, purpose-built transit, rather than occupation, is the predicted footprint, consistent with the null results of artefact SETI to date (Wright et al., 2022). Ivliev (2026) presses the opposite conclusion: once autonomous machine industry matures, outward expansion becomes too cheap and too useful for any civilization to refuse, and the attractor is quiet distributed expansion rather than concentration. We accept the premise that expansion is cheap, and we concede that for an undiscounted maximizer of total integrated computation Ivliev's inference goes through, since controlled mass grows as t^3. The disagreement localizes to the auxiliary assumptions of Section 4.5: under temporal discounting, latency-bound unified computation, or bounded planning horizons, expansion does not compete with concentration in yield per unit of controlled resource, and it is civilizations satisfying at least one of those conditions that the MTH captures. The two proposals are nonetheless observationally adjacent, since both predict a Galaxy free of conspicuous macro-engineering, and they diverge where the MTH is testable: Ivliev's filter predicts no privileged loci, while the MTH concentrates its residues on a short list of named systems.

The kugelblitz weak link. The only positive high-energy signature we predict (T6) inherits the controversy over micro-black-hole formation from radiation (Álvarez-Domínguez et al., 2024; Loeb, 2024). We therefore weight T6 as the cheapest test, not the strongest: its null closes a speculative channel; its detection would be extraordinary but stands or falls with Dvali and Osmanov (2023) physics, not with the MTH core.

Scope of the ω Cen bet. The MTH is a selection theory over a class; ω Cen is its highest-ranked accessible member, not its definition. Falsification at ω Cen (T3, or the ranking demotion of T4) retires the flagship application and substantially weakens the hypothesis (the Galaxy's best candidate failing is evidence about the class), but formally the theory survives in 47 Tuc, M54, and extragalactic nuclear clusters. We accept the methodological hazard this creates (unfalsifiability by retreat) and therefore bind ourselves to the class-level prediction: *if no Galactic globular-cluster IMBH with $a_\star \gtrsim 0.9$ exists, the MTH is wrong for the Milky Way*. Stating that in deliverable form matters, because a spin measurement requires an inspiral and the per-cluster EMRI rate is uncertain by orders of magnitude (Section 5.3). The settleable version is conditional on yield: if LISA returns spin measurements for ≥3 Galactic globular-cluster IMBHs over its mission and none has $a_\star \gtrsim 0.9$, we treat the class prediction as failed. A mission that returns fewer than three leaves the class claim unadjudicated rather than supported, and we commit in advance to reporting it that way, not to reading a null yield as survival.

8 Conclusion

We have argued, first as a matter of established physics, that rapidly spinning massive black holes in dense old stellar clusters constitute the most favourable present-day environments we identify for long-term computation, on the comparison set of Section 4 and with the cold-space alternative left open (by factors of ∼11–60 in energy per unit fuel against complete fusion, of order 10^2 or more only on the MAD $\eta > 1$ branch, $\sim 10^9$ in continuous power, a realized $\sim 10^3$ (fed flow) to $\sim 10^7$ (ambient) in entropy-disposal cost once delivery overheads are charged, conditional on carrier modes colder than the 2.7 K sky (of order unity otherwise) against the $\sim 10^{12}$ floor-to-floor ratio (Section 4.2), and unbounded factors in archival density), and second, as an explicitly speculative hypothesis, that this gradient acts on the oldest technological civilizations as an attractor whose operation would look, from outside, like the silence we observe. The Macro Transcension Hypothesis inherits the thermodynamic insight of transcension and aestivation while repairing their respective defects: it needs no exit from the universe and no dormancy, and it gives up nothing by not waiting, since Bennett et al. (2019) showed that waiting for a colder background does not lower the erasure cost, and it imports the cold reservoir into the present.

Its principal virtue is observational specificity. The hypothesis selects a named target (Omega Centauri, whose contested 8×10^3–$5 \times 10^4\ M_\odot$ central object, perfect electromagnetic silence, and 12-Gyr head start make it scientifically urgent on entirely conventional grounds) and commits to kill criteria that existing and funded instruments will adjudicate: LISA's resolution of the IMBH-versus-swarm question and, above all, its spin measurement; deep accretion and waste-heat limits from JWST and southern radio arrays; and a first-of-its-kind neutrino monitoring campaign with KM3NeT. If the central object proves to be a remnant swarm, a light point mass, or a slowly spinning hole (on the thin-disc spin-up branch; Paper E, Section 3.4), the hypothesis fails at its flagship and we have said so in advance. If instead the 2030s deliver a massive, gas-starved, near-extremally spinning black hole in the quietest old cluster in the sky (the thin-disc spin-up branch of Paper E, Section 3.4; a MAD-mode spin-up would end at $a_\star \approx 0.035$–0.07), the question this paper poses will merit serious attention.

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Data availability

No new observational data were generated. All cited measurements are available in the referenced publications.

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A Staged exploitation architecture

Epistemic status: This section is speculative civilizational engineering. It is included not as prediction but because the falsification program of Section 6 requires a concrete model of what would be observable at each stage. Each stage uses only physics established in Section 4 plus engineering extrapolations that are individually defended in the cited literature; the conjunction is unargued-for and carries the usual burden of compounded speculation.

We model the migration as five phases (Fig. 8), parameterized for a civilization departing a planetary system ~5 kpc from ω Cen.

Phase 1 — Scouts. Gram-scale beamed-sail probes at 0.1–0.2 c (Lubin, 2016), with photon-braking and magnetic-sail deceleration into the cluster halo; transit $\sim 10^5$ yr. Objectives: resolve the IMBH-versus-swarm question in situ, map the rocky-body inventory, and emplace a deceleration relay for following waves. *External signature: none detectable at interstellar range.*

Phase 2 — Self-replicating industry. Kilogram-scale seed factories bootstrap exponentially from asteroidal material, following the NASA self-replicating-systems concept (Freitas and Gilbreath, 1982); decades to centuries to 10^{3+} factory units and a relay laser. *Signature: waste heat of order early-industrial — far below detectability.*

Phase 3 — Compute swarm. The main payload (substrate-independent minds, by this stage the default cargo; cf. Smart 2012) arrives and assembles a free-flying compute swarm. While the hole is being fed, the swarm's inner working boundary sits at $\gtrsim 10^2\, r_g$ ($\sim 3 \times 10^6$ km at $2 \times 10^4\, M_\odot$): the companion infrastructure analysis (Swanson, 2026a) finds that accretion-flow erosion excludes hardware inside $\sim 30\, r_g$ of a fueled hole, and differential Lense–Thirring precession already excludes rigid formations at $10^2\, r_g$, so the swarm must be free-flying there in any case. Controlled star-lifting of brown dwarfs establishes a thin sub-Eddington disk; BZ extraction begins at low spin. Tiered orbits allocate clock rate (Section 4.4), though for an operating (fueled) system the stable dilation on offer at the $10^2\, r_g$ working boundary is modest, $d\tau/dt \simeq 0.985$; the deep-orbit clock rates of Fig. 6 become available only to dormant or archival elements around an unfueled hole. *Signature: faint, soft accretion luminosity — the one phase in which the system would be conventionally observable; at ω Cen, current limits already exclude an active Phase 3 fed above $\sim 4 \times 10^{-3}$ of the Bondi rate (Mahida et al., 2026) or radiating above $\sim 10^{-12}$ of Eddington (Haggard et al., 2013; Chen et al., 2026).*

Phase 4 — Spin-up. Sustained prograde feeding ($\sim 10^{4–5}$ stars over $3.4–5.3 \times 10^7$ yr, 0.79 Salpeter e-foldings to reach Thorne's radiation-limited $a_\star = 0.998$ ($M_f/M_i \approx 2.2$) (Bardeen, 1970; Thorne, 1974; Swanson, 2026a); Section 4.1) drives $a_\star \to 0.9–0.998$; the ISCO migrates inward by a factor ~5; the fueled swarm holds its $\sim 10^2\, r_g$ working boundary throughout (Swanson, 2026a). *Signature: episodic jet activity if imperfectly managed; high spin as a permanent record. Secular cluster-core depletion is a weaker observable than the 2-pc figure suggests: the accreted $2.4 \times 10^4\, M_\odot$ (Section 4.1) is 22.5 per cent of the stellar mass inside 2 pc but only 1.29 per cent of the mass inside the 10-pc reservoir the fuel census requires, recomputed directly against the enclosed-mass profile of Section 5.2 (Table 5), the removed stellar mass reappears as central point mass so the total enclosed mass profile is preserved. The stellar mass function is not, and that is the observable P4c names. The ~4-Gyr core relaxation time is the right refill clock for the 2-pc deficit ($\log t_{rc} = 9.60$) and the wrong one for the 10-pc reservoir the fuel census actually requires, whose half-mass relaxation time is $\log t_{rh} = 10.09$, or ~12 Gyr (Harris, 1996); the 10-pc deficit is therefore durable on the order of the cluster's age rather than a third of it (Swanson, 2026a, ; Appendix C).* We flag the strongest hidden premise of the whole architecture here: Phases 3–5 presuppose institutional or goal stability over 10^8 yr, longer than mammalian evolutionary history, and stellar feeding requires active orbit-shaping of individual stars with non-trivial delta-v budgets. We know of no principled argument that either is achievable; the architecture is offered conditional on both.

The delta-v can at least be priced to order of magnitude, and the pricing rules out two of the three options a reader might imagine. Delivering a star inward from a near-circular cluster orbit requires removing nearly all of its orbital angular momentum; the per-star impulse is of order the local orbital speed, tens of km s^{-1} across the central 1–10 pc and growing with the radius of the ∼10-pc reservoir that the fuel census of Section 3.1 requires. The energy is not the obstacle (binding energy is released on the way down); the angular momentum is.

Natural refilling does not supply the rate. Phase 4 needs $10^{4–5}$ stars over 3.4–5.3 × 10^7 yr, i.e. ∼1–2 × 10^{-3} stars yr^{-1} for the 6.8 × 10^4 stars the mass budget implies. Natural loss-cone refilling, which resonant relaxation drives near the influence radius (Rauch and Tremaine, 1996), delivers stars to the hole at the tidal-disruption rate, ∼10^{-8}–10^{-7} yr^{-1} for IMBHs in evolved clusters and ∼5 × 10^{-8} yr^{-1} in the ω Cen models of González Prieto et al. (2025) specifically (Swanson, 2026a). The shortfall is four to five orders of magnitude, and the repaired episode duration of Section 4.1 moves it by less than a factor of two. The natural channel is not a slower alternative to engineered delivery; it is not a fuel supply for this architecture at all, and essentially the whole Phase-4 budget must be actuated.

Beamed-momentum tugs are excluded, on the cluster's own light rather than on the fuel ceiling. The kinetic work is not the problem. Imparting Δv to the 2.4 × 10^4 $M_\odot$ that Phase 4 consumes costs $\frac{1}{2}M\Delta v^2$, which averaged over the 5.3×10^7 yr episode of Section 4.1 is 15 $L_\odot$ at 20 km s^{-1} and 3.7×10^2 $L_\odot$ at 10^2 km s^{-1}. Paying the same momentum in photons costs $E = \Delta p\, c$, larger by $2c/\Delta v$, which is a factor of 3 × 10^4 at 20 km s^{-1}.

A ceiling of 7.6 × 10^5 $L_\odot$ was previously described as "the JWST-limited accretion luminosity" and the tug priced against it (Appendix C). Two corrections are needed. First the label: 7.6 × 10^5 $L_\odot$ is the ambient Bondi accretion power at the fiducial mass and spin, taken from the companion infrastructure analysis (Swanson, 2026a); the JWST limit of test T1 is 2.5 × 10^{26} W, which is 0.65 $L_\odot$, and the two differ by a factor of 10^6 because they are different quantities. Second the verdict. Against the Bondi ceiling the exclusion is not robust: at 10^2 km s^{-1} the photon tug costs 2.2 × 10^6 $L_\odot$, or 2.9× the ceiling, but the impulse this section itself calls for is of order the local orbital speed, and the King profile of Section 5.2 gives 8.2, 15.2 and 28.3 km s^{-1} at 1, 2 and 10 pc. At 20 km s^{-1} the photon tug costs 4.5 × 10^5 $L_\odot$, which is 0.59 of the ceiling and therefore fits; the exclusion threshold sits at $\Delta v \gtrsim 34$ km s^{-1}, above the whole orbital-speed bracket.

The exclusion survives on a comparator that does not depend on the contested ceiling. The photons are absorbed by the star being moved and reradiated at stellar temperature, so they are added to the cluster's integrated light, and ω Cen radiates 1.09 × 10^6 $L_\odot$ in total at $M_{V,t} = -10.26$. The photon tug costs 17 per cent of that at 8.2 km s^{-1}, 41 per cent at 20, and 58 per cent at 28.3: a diffuse excess that ordinary photometry of the cluster excludes at every Δv in the bracket. The kinetic channel over the same bracket costs between 2 × 10^{-6} and 3 × 10^{-5} of the cluster's light. Gravitational-assist sequencing among cluster stars, which uses other stars as momentum reservoirs and pays only actuator overhead, is the one scheme that survives its own thermal signature; the margin behind this conclusion is three to four orders of magnitude.

Delivery geometry, not merely delivery rate, is what the spin fossil records. Loss-cone stars arrive isotropically, and isotropic feeding random-walks a hole's spin rather than raising it (Section 6; Swanson 2026a). Coherent prograde delivery is therefore not an efficiency improvement on the natural channel but the only channel that produces the observable this paper treats as its decisive test. A full account (which stars, in what order, at what actuator cost, in what orbital orientation) has not been produced and remains the largest quantitative gap in this section.

Phase 5 — Endpoint. Near-extremal spin; waste entropy disposed across the horizon (Section 4.2); residual stellar population dispersed or consumed. During fueled operation the swarm computes at the

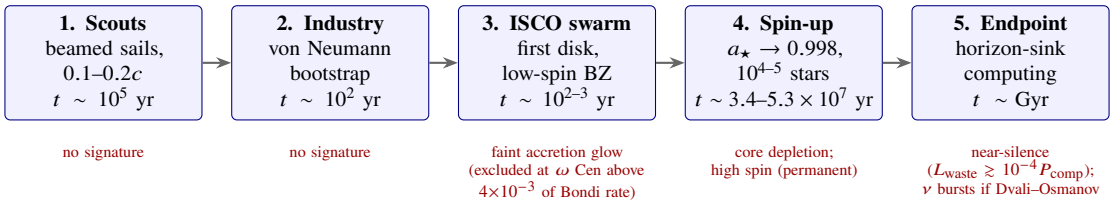


Figure 8: The five-phase exploitation architecture (speculative; see epistemic-status box) and the external signature of each phase. Only Phase 3 is conventionally luminous; the permanent observables are dynamical (spin, core depletion), which is why gravitational-wave spin measurement is the decisive test.

$\gtrsim 10^2\, r_g$ working boundary ($d\tau/dt \simeq 0.985$); if the system instead enters a dormant, unfueled archival mode, elements can descend to the near-ISCO orbits of Fig. 6 and bank the full ∼11:1 stable dilation. The system is nearly, but not perfectly, thermodynamically silent: transport losses in delivering waste entropy to the horizon impose a waste-heat floor $L_{\rm waste} \gtrsim 10^{-4}\, P_{\rm comp}$ (Swanson, 2026a), so current JWST mid-infrared limits already bound the computational throughput of any operating Phase-5 system at ω Cen to $P_{\rm comp} \lesssim 10^{3-4}\, L_\odot$; there is no radio leakage, and the horizon temperature is in the picokelvin range (Eq. 4). The sole conjectured emission channel is burst-mode: if intractable computations are delegated to manufactured micro black holes per Dvali and Osmanov (2023), their evaporation yields high-energy neutrino and gamma-ray transients. We note for completeness that the formation of micro black holes from focused radiation may be blocked by Schwinger-limit pair production (Álvarez-Domínguez et al., 2024; Breit and Wheeler, 1934; Schwinger, 1951); this conclusion is contested (Loeb, 2024), and the MTH does not depend on the channel — it affects only the strength of prediction T6 below.

B Pure-loss capacity of the delivery channel

This appendix derives the gains quoted in Section 4.2. The carrier is one photon of energy $E_{\rm carr} = h\nu_p$ at the plasma cutoff. The quantity wanted is the number of bits b it can bear, since the realized gain over the 2.7 K sky is $G = (k_{\rm B} T_\gamma \ln 2/E_{\rm carr})\, b$.

Capacity. For a bosonic channel of transmissivity η_k in mode k, the classical capacity is $C = \max \sum_k g(\eta_k N_k)$ bits per use, with $g(x) = (x+1)\log_2(x+1) - x \log_2 x$ and mean occupations N_k subject to $\sum_k \hbar\omega_k N_k = E$ (Giovannetti et al., 2004, Eqs. 4, 5, 14). The bound is attained by the Holevo quantity, which for pure-state ensembles is the entropy of the average state. One photon spread evenly over M modes has $N_k = 1/M$, and for $M \gg 1$ the capacity is $\log_2(eM)$ bits; with transmissivity η it is $\eta \log_2(eM/\eta)$. A one-wavelength packet is a single mode, for which $g(1) = 2$ bits, so the Bekenstein figure of 57 bits per wavelength is loose by a factor of ∼ 28 for the configuration named in Section 4.2. Bits per carrier go as the logarithm of the mode count, not as the packet length.

Mode count. A packet of coherence length L and bandwidth of one octave above the cutoff offers $M_{\rm t} = L/\lambda$ temporal modes. The spatial modes are the space-bandwidth product $M_{\rm sp} = A_t \Omega_c/\lambda^2$, with $\Omega_c = 27\pi r_g^2/(10^2 r_g)^2$ the capture solid angle and A_t the emitting aperture. For $M = 2\times10^4\, M_\odot$ and a fiducial aperture of diameter $12.3\, r_g$, $M_{\rm sp} = 1.8\times10^5$ at 4.3 kHz and 7.8×10^{13} at 90 MHz, since the count scales as λ^{-2}. Apertures of $4\, r_g$ and $400\, r_g$ move the realized gains by less than a factor of two.

Gains. The values in this paragraph assume modes in the vacuum state; the thermal case follows below. For $L = 100$ km the lossless capacity is 19.4 bits per carrier at 4.3 kHz and 61.0 at 90 MHz. Loss enters once, as the received occupation η/M, and the code rate enters as a separate factor, as in Table 2. The realized gains are 9.3×10^6 (ambient) and 1.3×10^3 (fed flow) on the conservative budget, 1.3×10^8 and 1.9×10^4 on the optimistic budget, and 1.8×10^8 and 2.7×10^4 lossless. Lengthening the packet to 3×10^3 km buys a factor 1.2 at the ambient floor, where the linear law of earlier drafts would have bought 30. Reaching $G = 10^9$ at the ambient floor needs $b = 109$ bits and hence $M \sim 10^{32.5}$ modes, and at the fed-flow floor $b = 2.3 \times 10^6$ bits; neither is reachable at any astrophysical length. The gains differ between floors by a factor of $\sim 7 \times 10^3$: the mode count adds 42 bits from the ambient to the fed floor while $k_B T_\gamma \ln 2 / E_{\mathrm{carr}}$ falls by 4.3 orders of magnitude.

Matter channel. A gram of matter at 100 km s^{-1} confined to 1 cm has kinetic energy 5×10^6 J and rest energy 9×10^{13} J. The Bekenstein capacity of 2.6×10^{38} bits is attained only by an object at the verge of collapse at that radius, about 6.7×10^{24} kg. The measured entropy of a solid is 5×10^{22} bits per gram for iron and 4×10^{23} for water ice at 300 K. Taking 10^{23} bits, the cost per bit is 5×10^{-17} J on kinetic charging and 2.7×10^{-10} J on ηmc^2 charging with $\eta = 0.30$, giving gains of 5×10^{-7} and $\sim 10^{-13}$. For a carrier in thermal equilibrium at T_c, $E/S = T_c$ and the gain is T_γ/T_c, so a matter carrier beats the CMB only if it is colder than 2.7 K, where its entropy is $\sim 10^{19}$ bits per gram and the channel at best breaks even.

Thermal background. The pure-loss capacity assumes a vacuum environment. The 2.7255 K sky has a mean occupation $N_{\mathrm{th}} = k_B T_\gamma / h\nu$ of 6.3×10^2 photons per mode at 90 MHz and 1.3×10^7 at 4.3 kHz. For a thermal-loss channel with transmissivity η the classical capacity of one photon spread over M modes (Giovannetti et al., 2014) is

$$C = M\,[g(\eta N + (1-\eta)N_{\mathrm{th}}) - g((1-\eta)N_{\mathrm{th}})]\,, \qquad N = 1/M, \tag{7}$$

which for $N_{\mathrm{th}} \gg 1$ reduces to $C \simeq \eta \log_2 e/[(1-\eta)N_{\mathrm{th}}]$ bits. With the conservative transmissivity $\eta = 0.089$ (capture 0.30, swarm 0.30, lapse 0.99) a carrier bears 1.1×10^{-8} bits at 4.3 kHz and 2.2×10^{-4} bits at 90 MHz. With the code rate of 0.5 the realized gain is 0.049 at both floors, because N_{th} scales as $1/\nu$ and cancels against the carrier energy in G. The optimistic budget ($\eta = 0.80$, code rate 0.9) gives 3.6 at both floors. The Clausius inequality $dS \le dE/T_\gamma$ gives the same conclusion: carriers thermalized at the sky temperature cannot export entropy at an energy cost per bit far below $k_B T_\gamma \ln 2$ (a factor of a few at most on the optimistic budget). The vacuum-state gains of the previous paragraph therefore hold only if the carrier modes remain colder than 2.7 K from the emitter to the horizon. That requires an aperture that occludes the sky, so that the modes the emitter fills are not coupled to it, and a path not thermalized by plasma at $T_e \gg 2.7$ K.

C Revision record

This appendix lists, per version, the changes that earlier drafts of the main text carried as running commentary.

v2.13 (2026-10-23). Numerical fixes from the farm audits D271 and D253. The Phase-4 star count in the natural-refilling paragraph is 6.8×10^4 (the rest-mass figure of the spin-up discussion; 5.6×10^4 is the hole increment), and the rate of ~ 1–2×10^{-3} stars yr^{-1} is unchanged. The expected false neutrino multiplet count is 6.6×10^{-12} per decade from $\mu^3/6 = 2.1 \times 10^{-17}$ per window and 3.16×10^5 windows

(was $\sim 10^{-11}$). Figure 2 is regenerated with the LISA reach anchored at the T4 threshold of $8.2 \times 10^3\ M_\odot$ as its caption states (was 5×10^3). The dashed extension of the efficiency curve in Figure 3 passes through $\eta(0.9995) = 0.3555$ (was 0.3597). The MAD-endpoint spin range is described as within 1σ of the natural-capture null of Paper E (was "at or below").

v2.12 (2026-10-22). The realized entropy-disposal gain of the radiative channel is corrected from 10^6–10^9 to about 10^3 in a fed flow (1.3×10^3 at the 90 MHz floor) and about 10^7 in the ambient medium (9.3×10^6 at 4.3 kHz), on the conservative budget of Table 2. The earlier range treated the Bekenstein bound as an achievable capacity; the pure-loss bosonic capacity (Giovannetti et al., 2004) grows as the logarithm of the mode count, so the length law $L = Ghc/(57\, k_{\rm B} T_\gamma \ln 2)$, the claim that the required length is common to all floors, and the abstract's coherence lengths of 0.2 to 3×10^3 km are withdrawn. Table 2 now carries the Bekenstein ceiling and the achievable gains side by side. The matter-channel gains (74, 247 and the quoted $\sim 10^2$) are struck: the physical entropy of a gram is 10^{22} to 10^{24} bits, which gives about 5×10^{-7} with kinetic charging and about 10^{-13} with opportunity-cost charging. The derivation is in the new Appendix B. The gains of $\sim 10^3$ and $\sim 10^7$ hold only for carrier modes kept colder than the 2.7 K sky from emitter to horizon; with thermally occupied modes (occupations 6.3×10^2 at 90 MHz and 1.3×10^7 at 4.3 kHz) the thermal-loss capacity gives a conservative gain of ~ 0.05 at both floors, and every statement of the larger gains carries that condition. The abstract, Section 4.2, the cost comparison with the Slysh halo, the energy-and-cost paragraph and the conclusion are updated to match.

v2.11 (2026-10-22). The attribution of Bennett et al. (2019) is corrected in the introduction, the aestivation paragraph, the corollaries, the Landauer discussion and the conclusion: they show that the erasure cost aestivation defers can be paid at any epoch and is not improved by waiting for a colder background. The argument that free energy not collected promptly is lost (stars burn regardless) is now stated as this paper's own, not Bennett et al.'s.

v2.10 (2026-10-20). Branch labels added at the spin-up discussion, the T5 row and its decision-tree node, and the conclusion: the near-extremal endpoint is the thin-disc spin-up branch of Paper E Section 3.4, and a MAD-mode spin-up would end at $a_\star \approx 0.035$–0.07, so T5 is branch-conditional. "MAD episodes" removed from Phase 4. Claim 1 and the conclusion softened from "wide margin" and "global optimum" to the comparison set of Section 4 and the ∼11–60 per-unit-fuel figure, consistent with the statement in Section 4.2 that the preference between architectures is not settled. The BH2 footnote now states the semi-major-axis residual using $P \propto a^{3/2}$ (∼8 per cent). The 21,100 $M_\odot$ entry of the mass table is labelled acceleration-model-informed. Draft-history commentary moved here from the body prose.

Earlier versions (corrected results retained in the text).

- *Spin-up duration.* Versions before the repaired integrator printed $\sim 10^8$ yr for the Phase-4 episode; the figure came from a spin-up integrator carrying four compounding errors and is withdrawn. Details: The shipped integrator (`figs/fA_calc7_v1.py`) evaluated z_2 as $\sqrt{3z_1^2 + a^2}$ rather than the $\sqrt{3a^2 + z_1^2}$ of Bardeen et al. (1972), took the + root of $r_{\rm ISCO} = 3 + z_2 \mp \sqrt{(3 - z_1)(3 + z_1 + 2z_2)}$, which is the retrograde orbit, omitted the $1/E_{\rm ISCO}$ in $\mathrm{d}a_\star/\mathrm{d}\ln M = (\ell - 2a_\star E)/E$, and integrated the duration against $\mathrm{d}m$ rather than $\mathrm{d}\ln m$. The first two put $\eta(0)$ at 0.0505 and $\eta(0.998)$ at 0.0515, so the figures and tabulated efficiencies in this paper were not produced by that code, which is how the error survived. The repaired routine returns $\eta(0) = 0.05719$, $\eta(0.998) = 0.32099$,

$r_{\mathrm{ISCO}}(0.998) = 1.23697\, r_g$, reaches $a_\star = 0.998$ at $M_f/M_i = 2.20240$, the growth factor quoted above, and reproduces the $\sqrt{6}$ of Bardeen (1970) as $a_\star \to 1$. A regression gate now pins all five.

- *Decision-tree mass split.* An earlier version split the mass branches at 5×10^3 on one side and 10^4 on the other, leaving masses in $[5 \times 10^3, 10^4)$, which contains the kinematic bound, with no terminal. The tree now splits at $8.2 \times 10^3\ M_\odot$.

- *Fusion denominators.* Earlier versions applied the hydrogen fraction X_{H} to the stellar yield alone; the Dyson factors then read ~80–420 rather than ~115–600.

- *Core radius.* The P2 table row printed 0.10 pc, which corresponds to $\sigma_c = 20.7\ \mathrm{km\,s^{-1}}$; the caption's convention $\sigma_c = 16.8\ \mathrm{km\,s^{-1}}$ gives 0.15 pc.

- *Bekenstein-bound length.* An earlier version stated the headroom property of the packet-length expression backwards; the text now states that the packet sits on the bound at every G and floor.

- *Delivery budget.* An earlier version claimed the 10^6–10^9 range was reached with capacity to spare; pricing the losses removed that claim, and v2.12 replaces the range itself.

- *Matter carriers.* The pellet-carrier accounting initially had no like-for-like treatment of rest energy; an intermediate draft gave 74 and 247, and v2.12 withdraws both in favour of the entropy-limited figures of Section 4.2.

- *Duration columns.* The enclosed-mass table (Table 5) once printed only the Eddington-years column (column 4); the elapsed-time column 5 was added.

- *Natural-spin channels.* Earlier versions adopted the coherent and the chaotic-accretion priors at once, which left the high-spin branch non-discriminating; the text adopts the coherent-growth prior, and the chaotic channel's weight at and below the T5 trigger had been placed in a small tail.

- *Refill clock.* The Phase-4 core-depletion discussion once used the ~4-Gyr core relaxation time for the 10-pc reservoir; the half-mass relaxation time of ~12 Gyr applies, which makes the deficit more durable and runs against the hedge drawn from the shorter clock.

- *Beamed-momentum tugs.* The exclusion was priced against $7.6 \times 10^5\ L_\odot$ labelled "the JWST-limited accretion luminosity" at an assumed $10^2\ \mathrm{km\,s^{-1}}$; the label is corrected to the ambient Bondi ceiling, and the exclusion now rests on the cluster's integrated light. The conclusion is unchanged; its margin was previously stated as a factor of a few.